

Section VI. CRUISE

7-25. CRUISE.

The cruise charts (Figure 7-13, sheets 1 through 61) show the torque pressure and fuel flow required for level flight at various pressure altitudes, airspeed, and gross weights. The cruise charts are presented for a clean drag configuration. However, any desired drag configuration (Figure 7-14) authorized configurations) can be solved by using the difference between the drag for the desired configuration and the clean configuration. This solution is ratioed to the drag for 5 square foot equivalent flat plate drag area. Add or subtract the calculated Δ % if the desired configuration to the clean configuration to solve the total torque required for the desired configuration. Then read the fuel flow for the desired configuration.

Thus, the effects of different drag resulting from changing configurations can be determined by applying the Δ F-SQ FT from the Drag Chart (Figure 7-14) to the cruise charts (Figure 7-13).

7-26. USE OF CHARTS.

DELETED

The primary use of the charts is illustrated by the examples provided in Figure 7-13. The first step for chart use is to select the proper chart, pressure altitude and anticipated free air temperature, (paragraph 7-2). Normally, sufficient accuracy can be obtained by selecting the chart nearest to the planned cruising altitude and FAT, or the next higher altitude and FAT. If greater accuracy is required, interpolation between altitude and/or temperatures will be required (Figure 7-14, sheet 1, example A). You may enter the charts on any side: TAS, IAS, torque pressure, or fuel flow, and then move vertically or horizontally to the gross weight, then to the other three parameters. Maximum performance conditions are determined by entering the chart where the maximum range of maximum endurance and rate of climb lines intersect the appropriate gross weight; then read airspeed, fuel flow, and torque pressure. For conservatism, use the gross weight at the beginning of cruise flight. For greater accuracy on long flights it is preferable to determine cruise information for several flight segments in order to allow for decreasing fuel weight (reduced gross weight). The following parameters

contained in each chart are further explained as follows:

a. *Airspeed.* True and indicated airspeeds are presented at adjacent sides of each chart. On any chart, indicated airspeed can be directly converted to true airspeed (or vice versa) by reading IAS vs TAS line. Maximum permissible airspeed (V_{NE}) limits appear on some charts. If no line appears, V_{NE} is above the limits of the chart.

b. *Torque Pressure (%).* Since pressure altitude and temperature are fixed for each chart, torque % required varies according to gross weight and airspeed.

c. *Fuel Flow.* Fuel flow is shown as a function of torque %. On any chart, torque % may be converted directly to fuel flow without regard for other chart information. All fuel flow information is presented with bleed air heater and de-ice off. Add two percent fuel flow (about 14 lb/hr) for bleed air heater on and increase fuel flow three percent (approximately 21 lb/hr) for anti-ice on. If both are operating, add five percent fuel flow (about 35 lb/hr) to chart values.

d. *Maximum Range.* The maximum range lines indicate the combinations of weight and airspeed that will produce the greatest flight range per gallon of fuel under zero wind conditions. When a maximum range condition does not appear on a chart it is because the maximum range speed is beyond the maximum permissible speed (V_{NE}); in such cases, use (V_{NE}) cruising speed to obtain maximum range.

e. *Maximum Endurance and Rate of Climb.* The maximum endurance and rate of climb lines indicate the airspeed for minimum torque pressure required to maintain level flight for each gross weight, FAT and pressure altitude. Since minimum torque pressure will provide minimum fuel flow, maximum flight endurance will be obtained at the airspeed indicated.

7-27. CONDITIONS.

The cruise charts are based on operation of 324/rotor/6600 engine RPM with de-ice and heater off.

EXAMPLE

WANTED

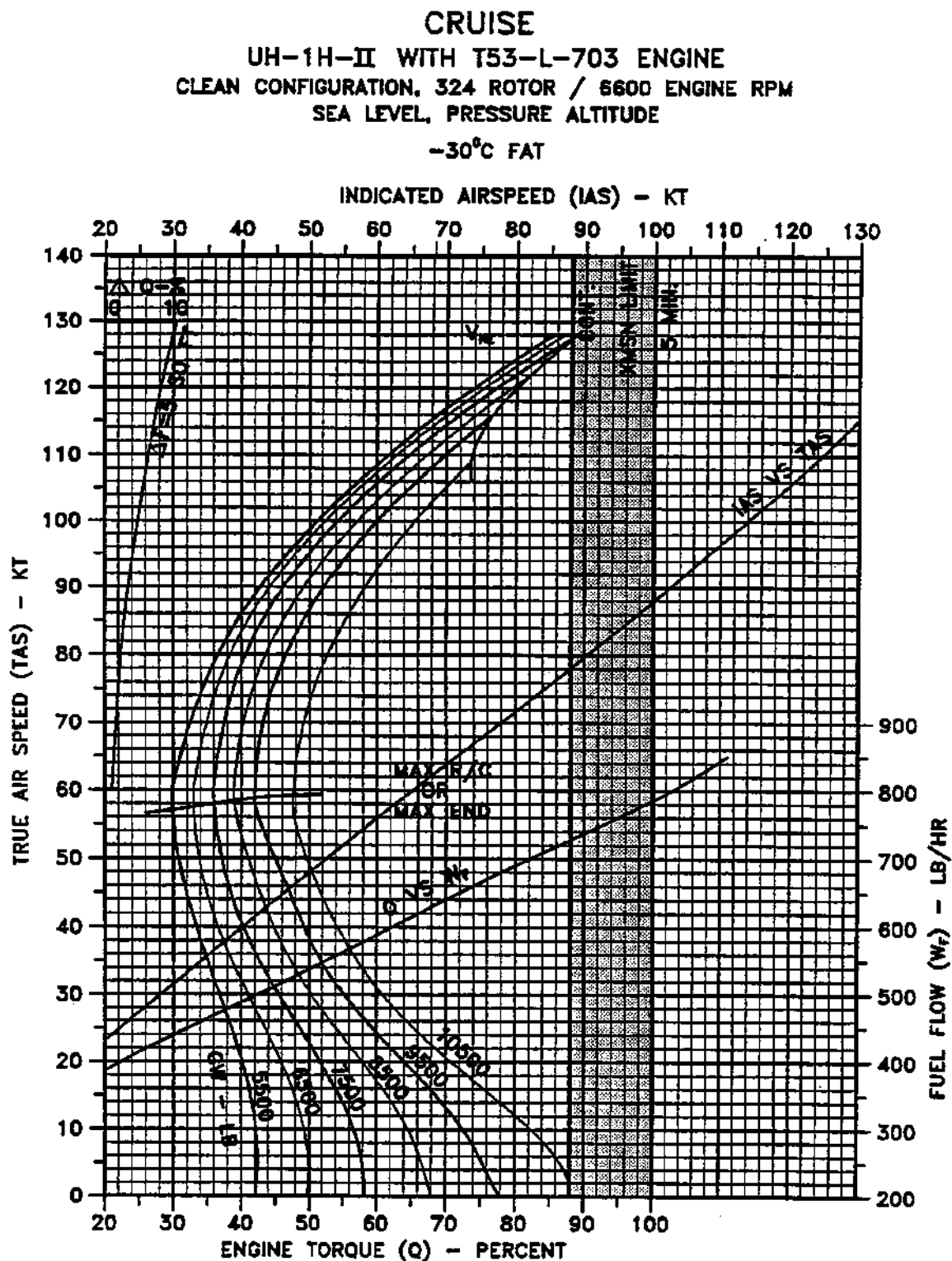
SPEED FOR MAXIMUM RANGE,
CALIBRATED ENGINE TORQUE REQUIRED AND FUEL FLOW AT
MAXIMUM RANGE SPEED AND FOR MAXIMUM ENDURANCE

KNOWN

CLEAN CONFIGURATION
PRESSURE ALTITUDE = SEA LEVEL, FAT = -30°C
GROSS WEIGHT = 9500 POUNDS

METHOD

LOCATE (-30°C , SEA LEVEL) CHART
FIND INTERSECTION OF 9500 POUND GROSS WEIGHT LINE WITH THE
MAXIMUM RANGE LINE, SAME AS THE V_{NE} LINE
TO READ SPEED FOR MAXIMUM RANGE (SAME AS V_{NE} SPEED):
 MOVE LEFT TO READ TAS = 115.0 KNOTS, MOVE RIGHT
 TO IAS VS TAS LINE AND MOVE UP TO READ
 IAS = 129.7 KNOTS
TO READ FUEL FLOW REQUIRED:
 MOVE DOWN TO THE Q VS W_f LINE TO READ
 FUEL FLOW = 670 LB/HR
TO READ CALIBRATED ENGINE TORQUE REQUIRED:
 MOVE DOWN TO READ ENGINE TORQUE = 76.0 %Q
FIND INTERSECTION OF 9500 POUND GROSS WEIGHT LINE WITH
THE MAX. END. LINE
TO READ SPEED FOR MAXIMUM ENDURANCE:
 MOVE LEFT TO READ TAS = 58.7 KNOTS
 MOVE RIGHT TO THE IAS VS TAS LINE, MOVE UP
 TO READ IAS = 63.5 KNOTS



UH-1H-II-FM-7-15-1

Figure 7-13. Cruise chart, clean configuration, -30°C, sea level, (Sheet 1 of 61)

EXAMPLE

WANTED

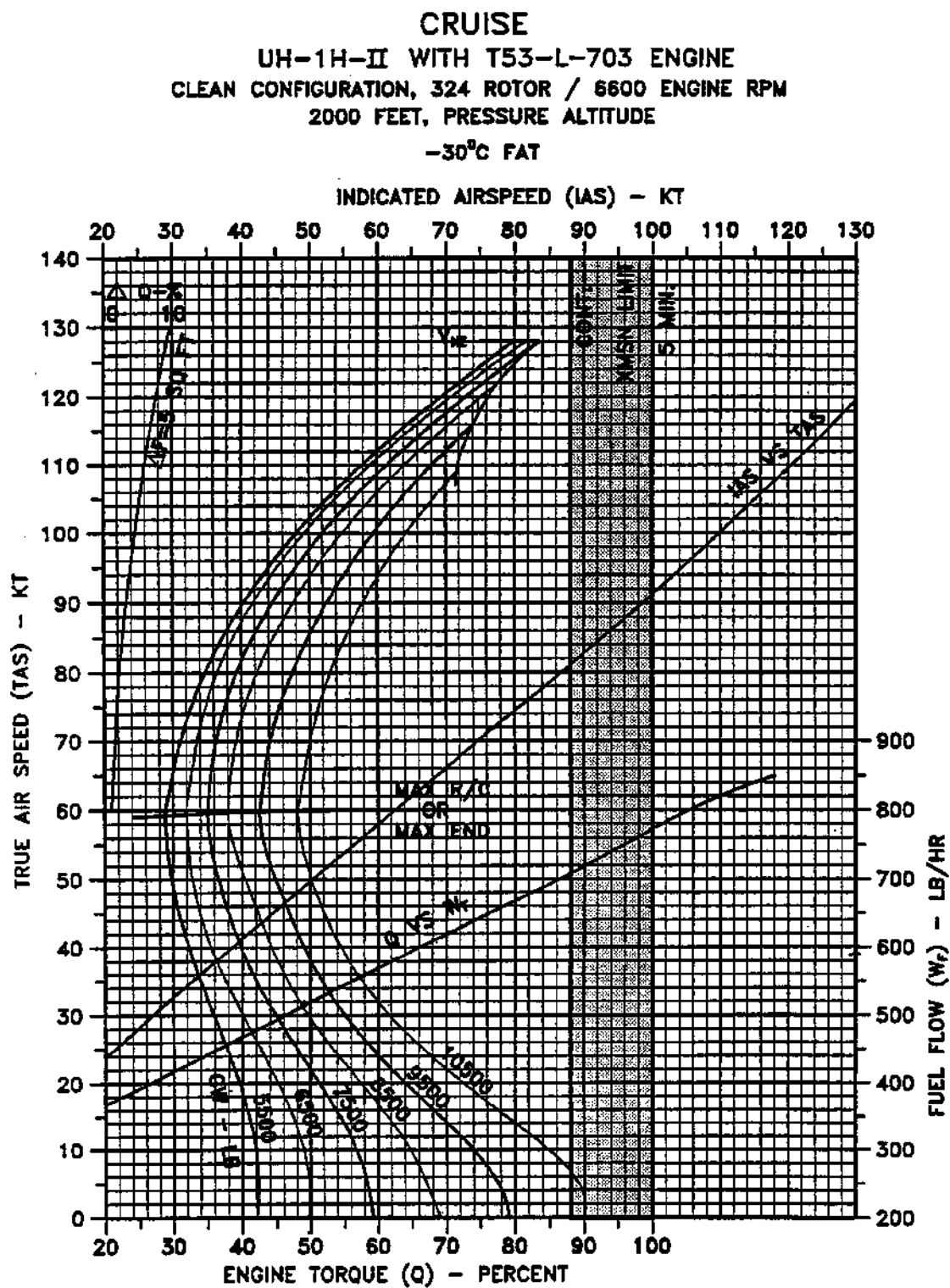
ADDITIONAL CALIBRATED TORQUE REQUIRED AND
FUEL FLOW FOR EXTERNAL DRAG CONFIGURATION

KNOWN

ΔF FOR EXTERNAL DRAG CONFIGURATION (FROM FIGURE 7-7)
= 4 SQUARE FEET
PRESSURE ALTITUDE = 2000 FEET, $FAT = -30^{\circ}C$
GROSS WEIGHT = 8500 POUNDS
TRUE AIRSPEED = 110 KNOTS

METHOD

ENTER TRUE AIRSPEED AT 110 KNOTS. MOVE RIGHT TO
8500 POUND GROSS WEIGHT LINE. MOVE DOWN TO THE
 Q VS W_F LINE. FROM THIS INTERSECTION, MOVE RIGHT
TO THE FUEL FLOW SCALE TO READ 588 LB/HR. FROM THE
ABOVE INTERSECTION MOVE DOWN TO THE ENGINE TORQUE
SCALE TO READ 64.0 %Q.
AGAIN ENTER TRUE AIRSPEED AT 110 KNOTS. MOVE RIGHT
TO THE 5 SQ.FT. ΔF LINE. MOVE UP TO READ 6.0 %Q.
DIVIDE 4 SQ.FT. BY 5 SQ.FT. = 80%. 80% OF 6.0 %Q =
4.8 %Q. ADD 4.8 AND 64.0 = 68.8 %Q. MOVE UP FROM ENGINE
TORQUE SCALE AT 68.8 %Q TO THE FUEL FLOW LINE
AND MOVE RIGHT TO READ 614 LB/HR.



UH-1H-II-FM-7-15-2

Figure 7-13. Cruise chart, clean configuration, -30°C, 2000 feet, (Sheet 2 of 61)

EXAMPLE**WANTED**

CALIBRATED ENGINE TORQUE REQUIRED FOR LEVEL FLIGHT,
FUEL FLOW, INDICATED AIRSPEED

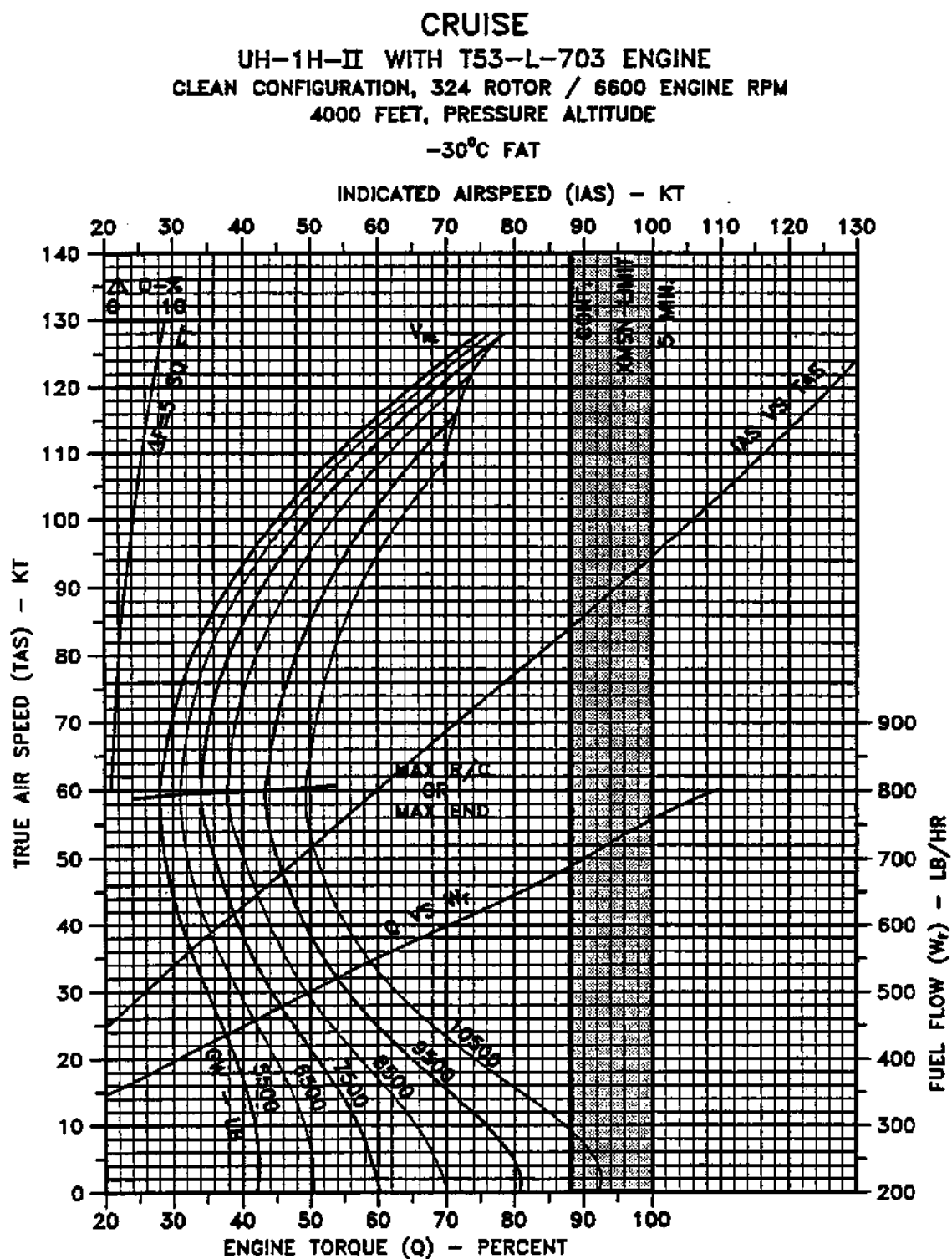
KNOWN

CLEAN CONFIGURATION
PRESSURE ALTITUDE = 5000 FEET, FAT = -30°C
GROSS WEIGHT = 9500 POUNDS
DESIRED TRUE AIRSPEED = 100 KNOTS

METHOD

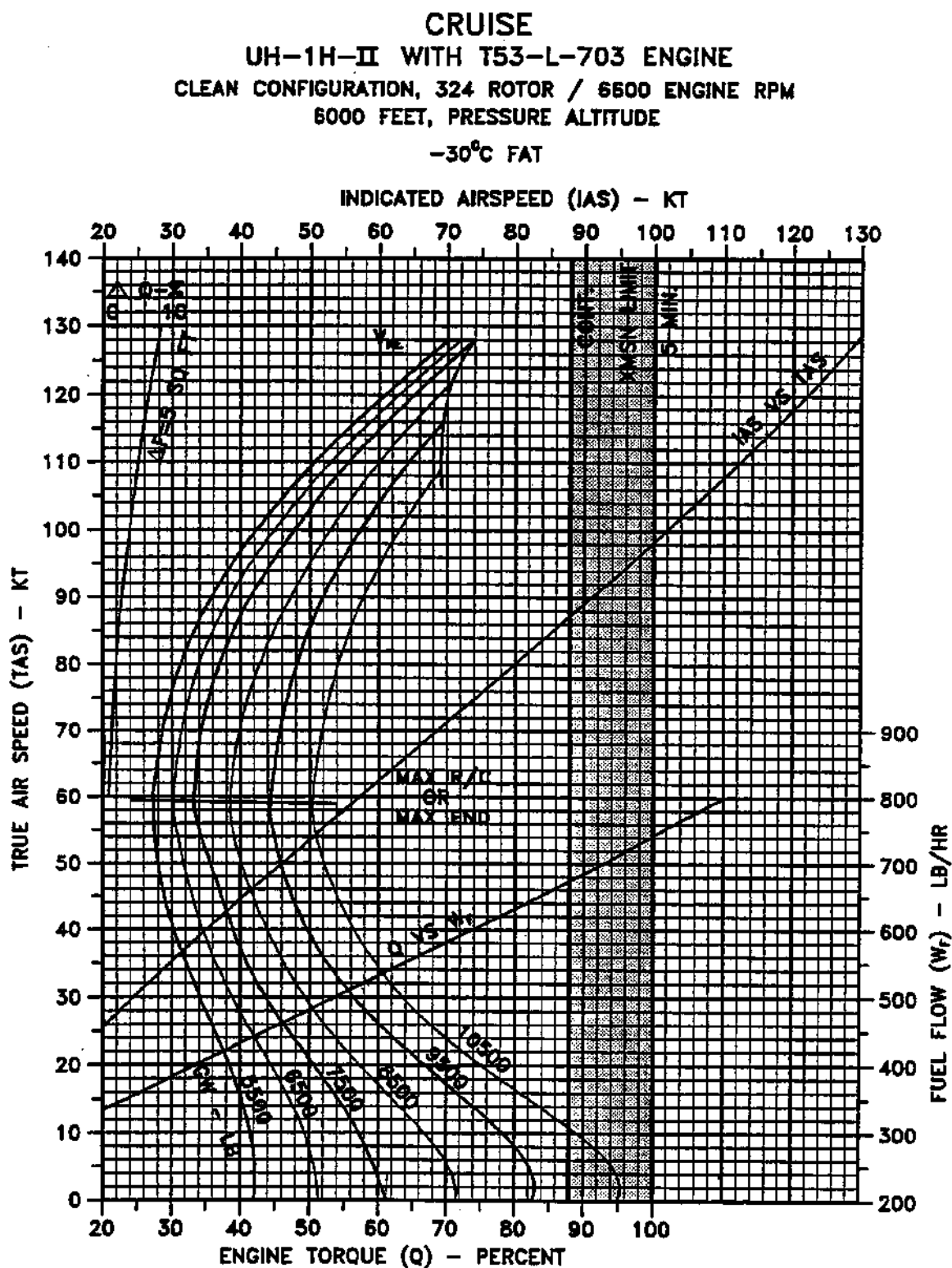
ENTER TRUE AIRSPEED AT 100 KNOTS
READ CALIBRATED ENGINE TORQUE, FUEL FLOW AND IAS ON EACH
ADJACENT ALTITUDE AND/OR FAT, THEN INTERPOLATE
BETWEEN ALTITUDE AND/OR FAT.

ALTITUDE, FEET	4000	6000	5000
FAT, $^{\circ}\text{C}$	-30	-30	-30
CALIBRATED TORQUE, %Q	58.5	57.4	57.9
FUEL FLOW, LB/HR	545	520	532.5
IAS, KNOTS	106	102	104



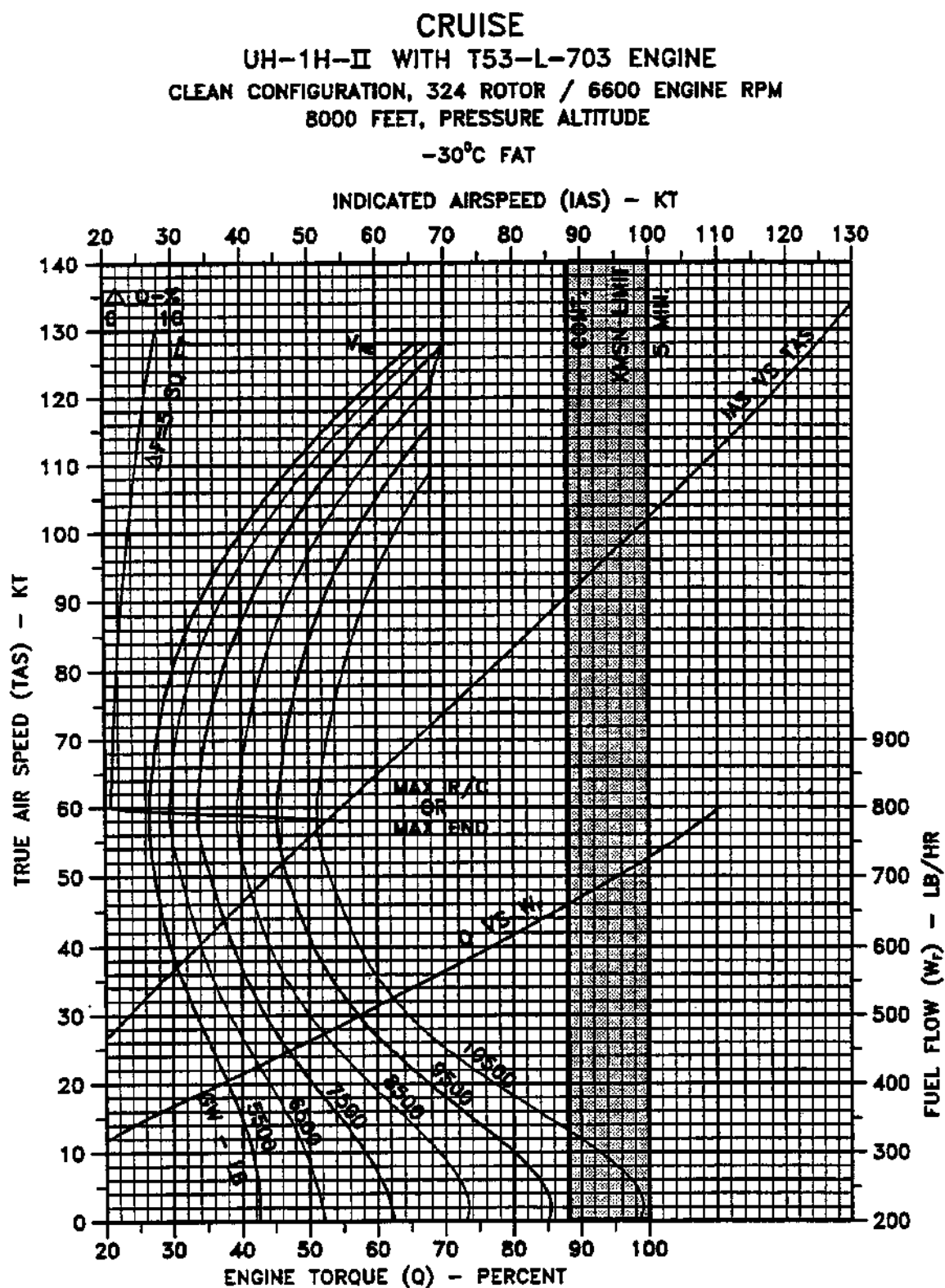
UH-1H-II-FM-7-15-3

Figure 7-13. Cruise chart, clean configuration, -30°C, 4000 feet, (Sheet 3 of 61)



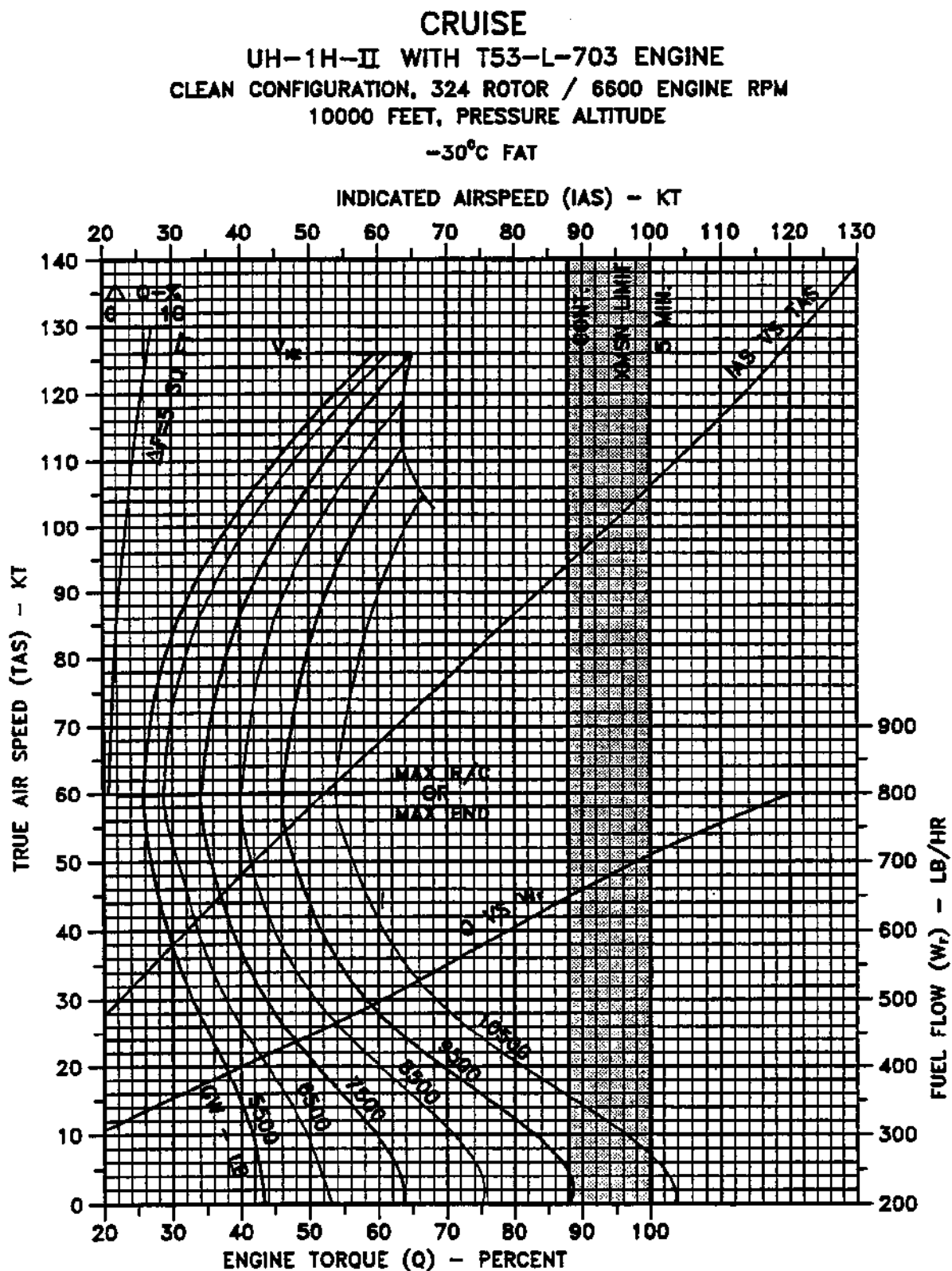
UH-1H-II-FM-7-15-4

Figure 7-13. Cruise chart, clean configuration, -30°C, 6000 feet, (Sheet 4 of 61)



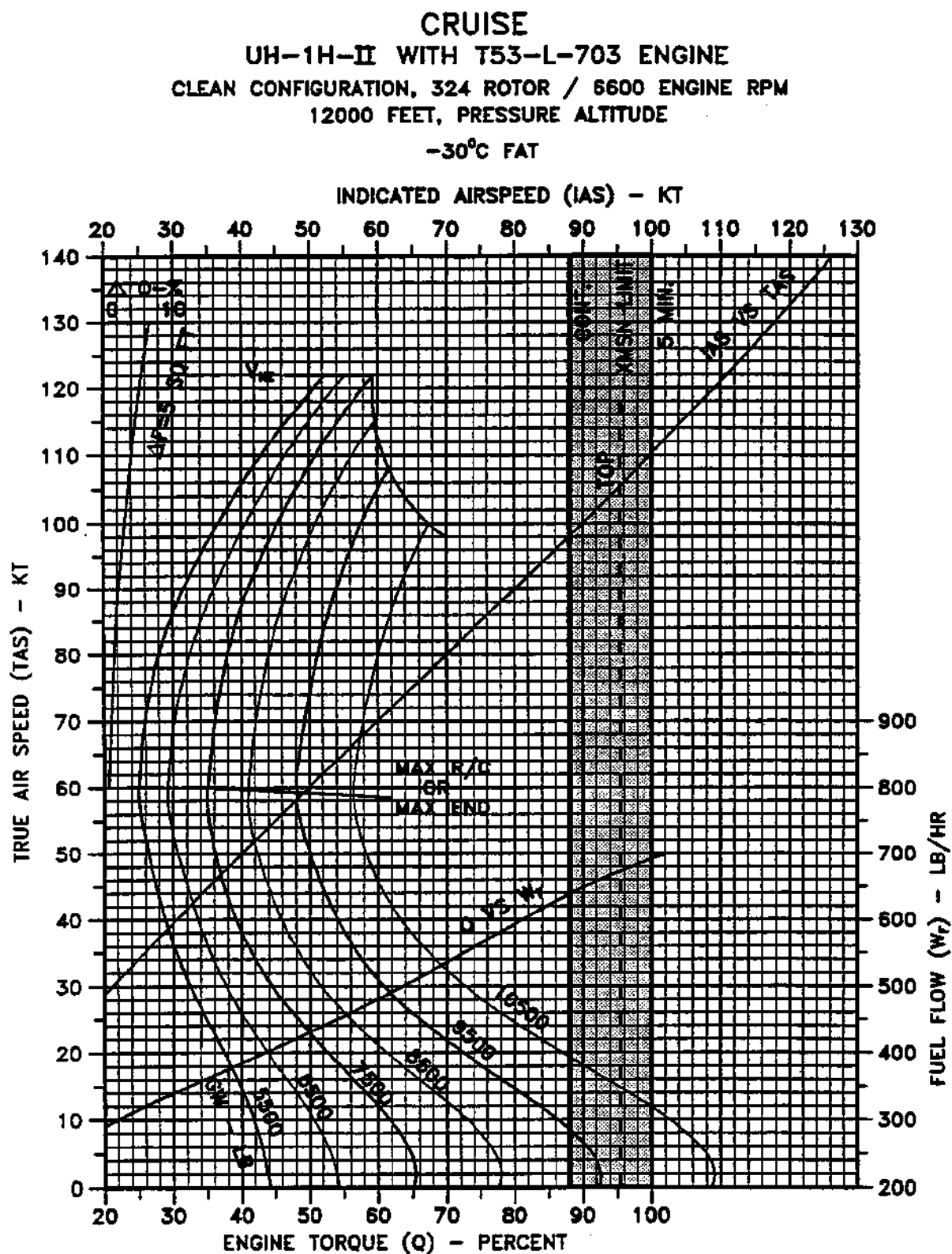
UH-1H-II-FM-7-15-5

Figure 7-13. Cruise chart, clean configuration, -30°C, 8000 feet, (Sheet 5 of 61)



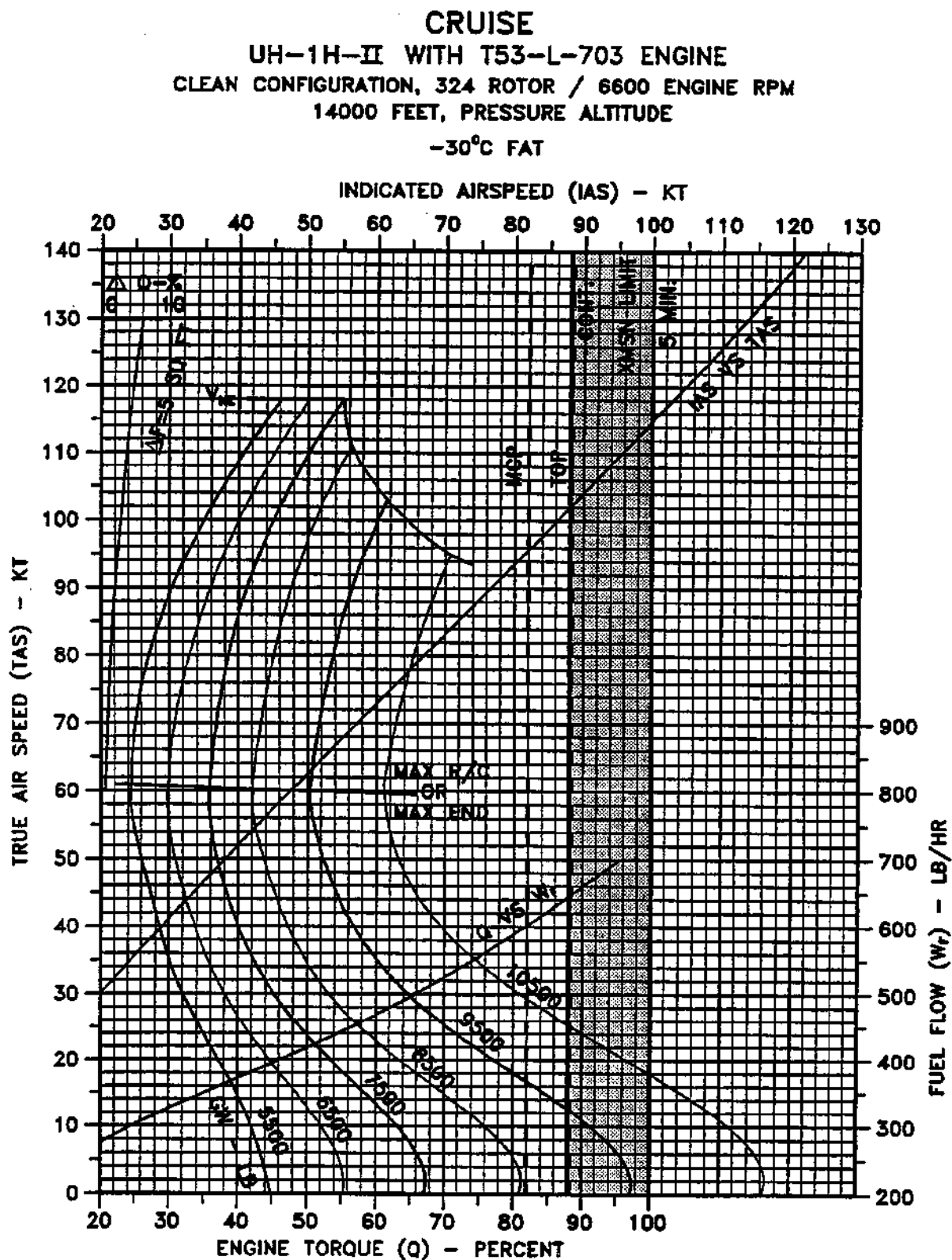
UH-1H-II-FM-7-15-6

Figure 7-13. Cruise chart, clean configuration, -30°C, 10000 feet, (Sheet 6 of 61)



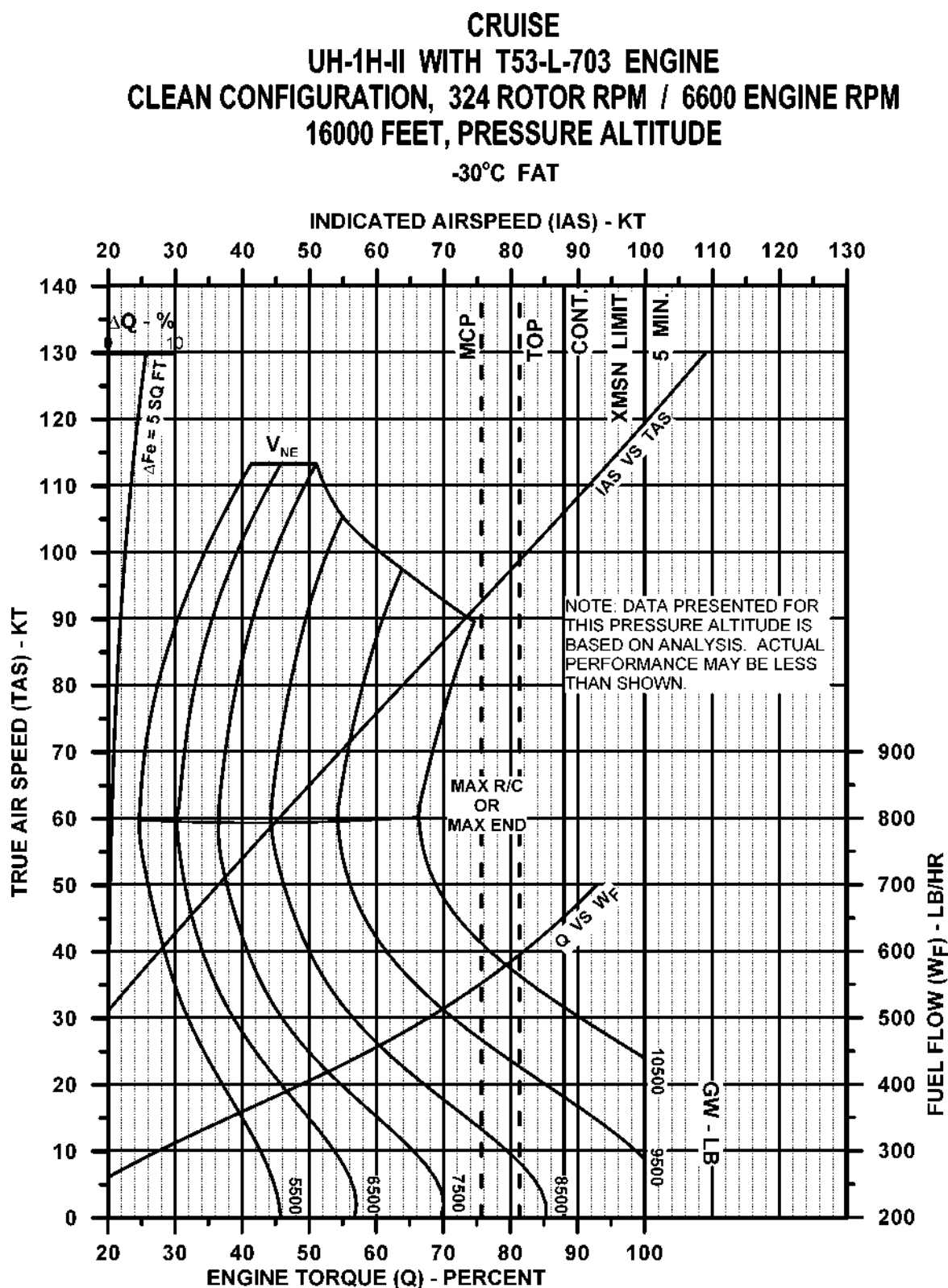
UH-1H-II-FM-7-15-7

Figure 7-13. Cruise chart, clean configuration, -30°C, 12000 feet, (Sheet 7 of 61)



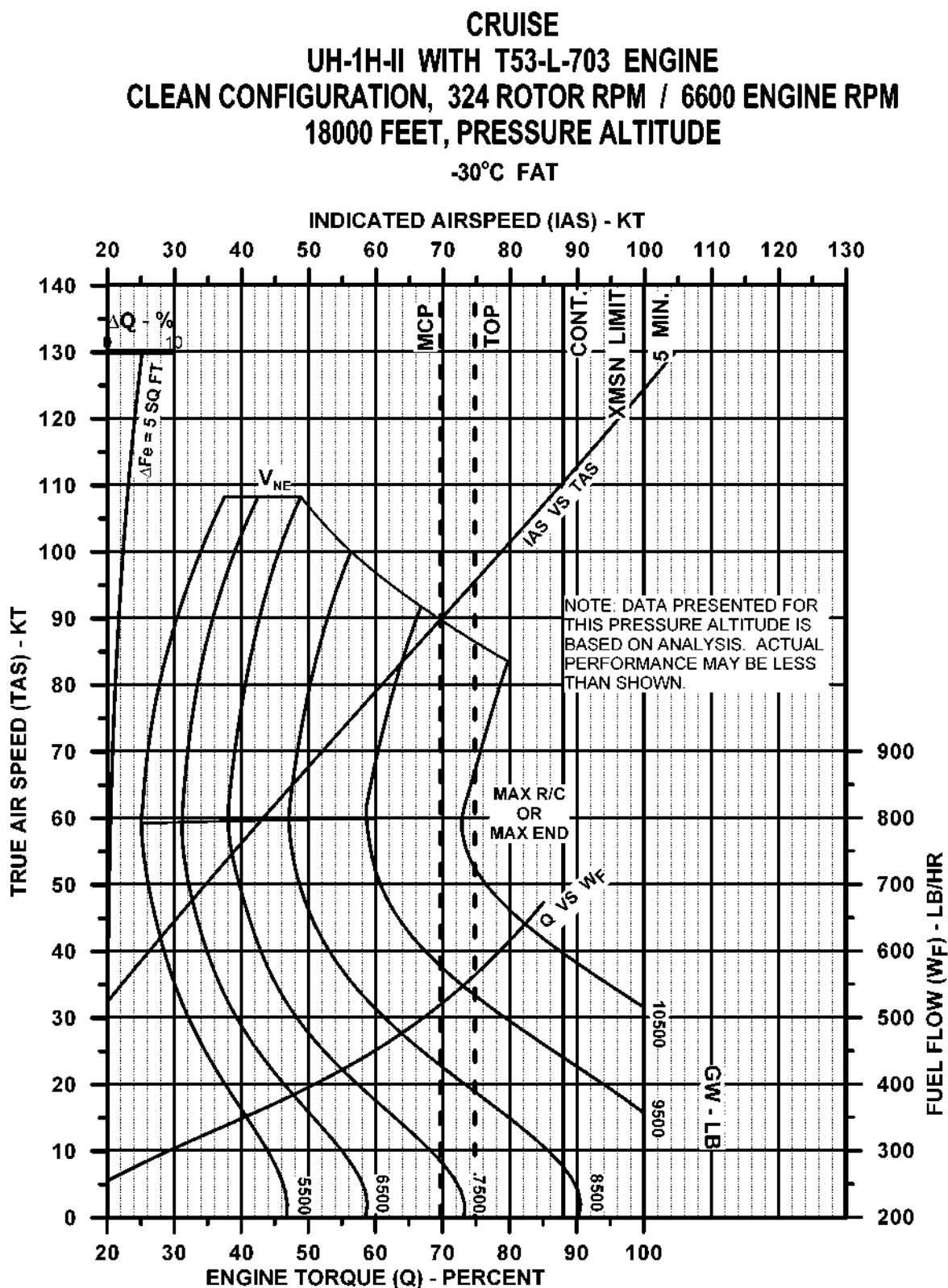
UH-1H-II-FM-7-15-8

Figure 7-13. Cruise chart, clean configuration, -30°C, 14000 feet, (Sheet 8 of 61)



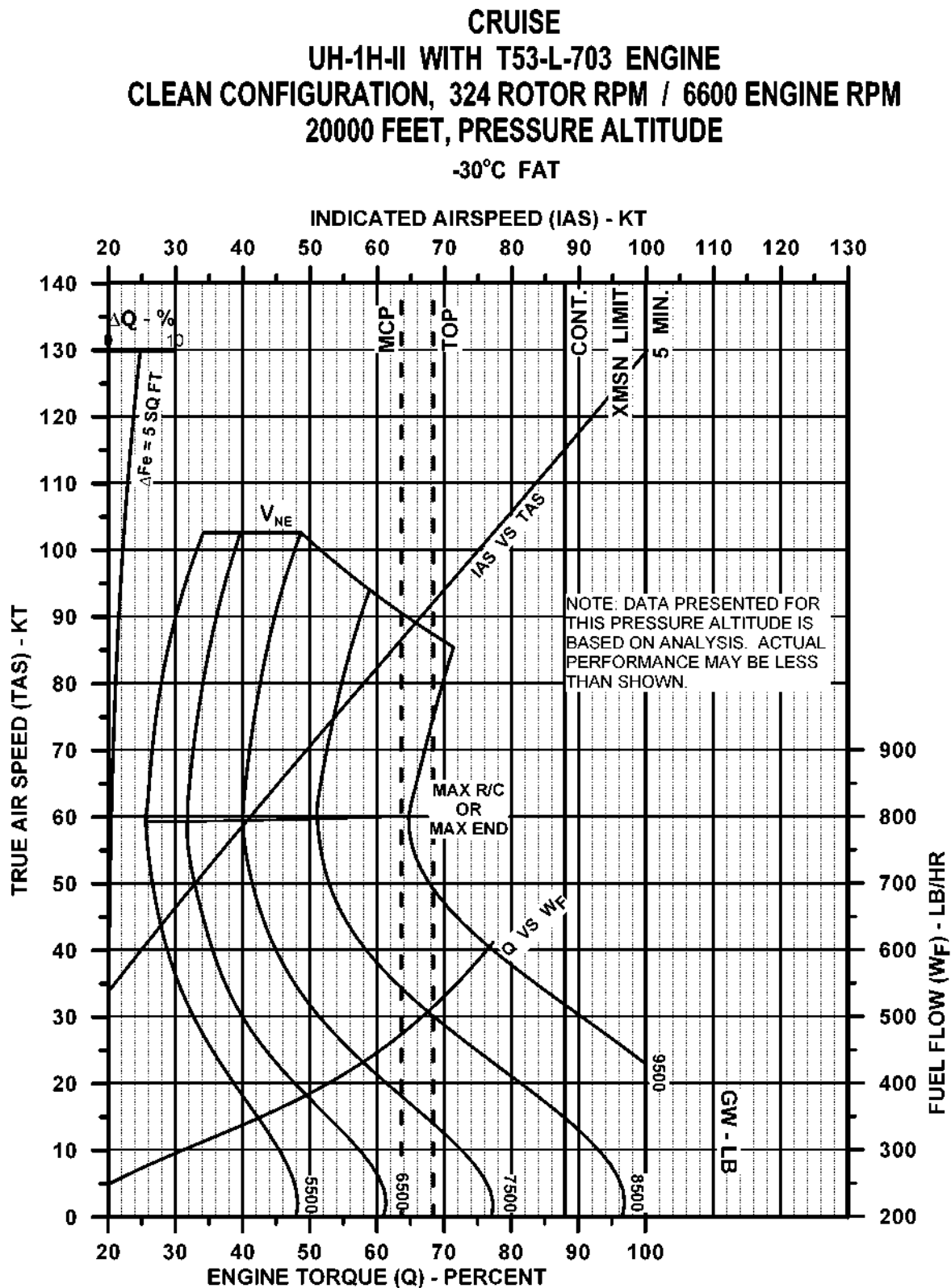
UH-1H-II-FM-7-15-9

Figure 7-13. Cruise chart, clean configuration, -30°C, 16000 feet, (Sheet 9 of 61)



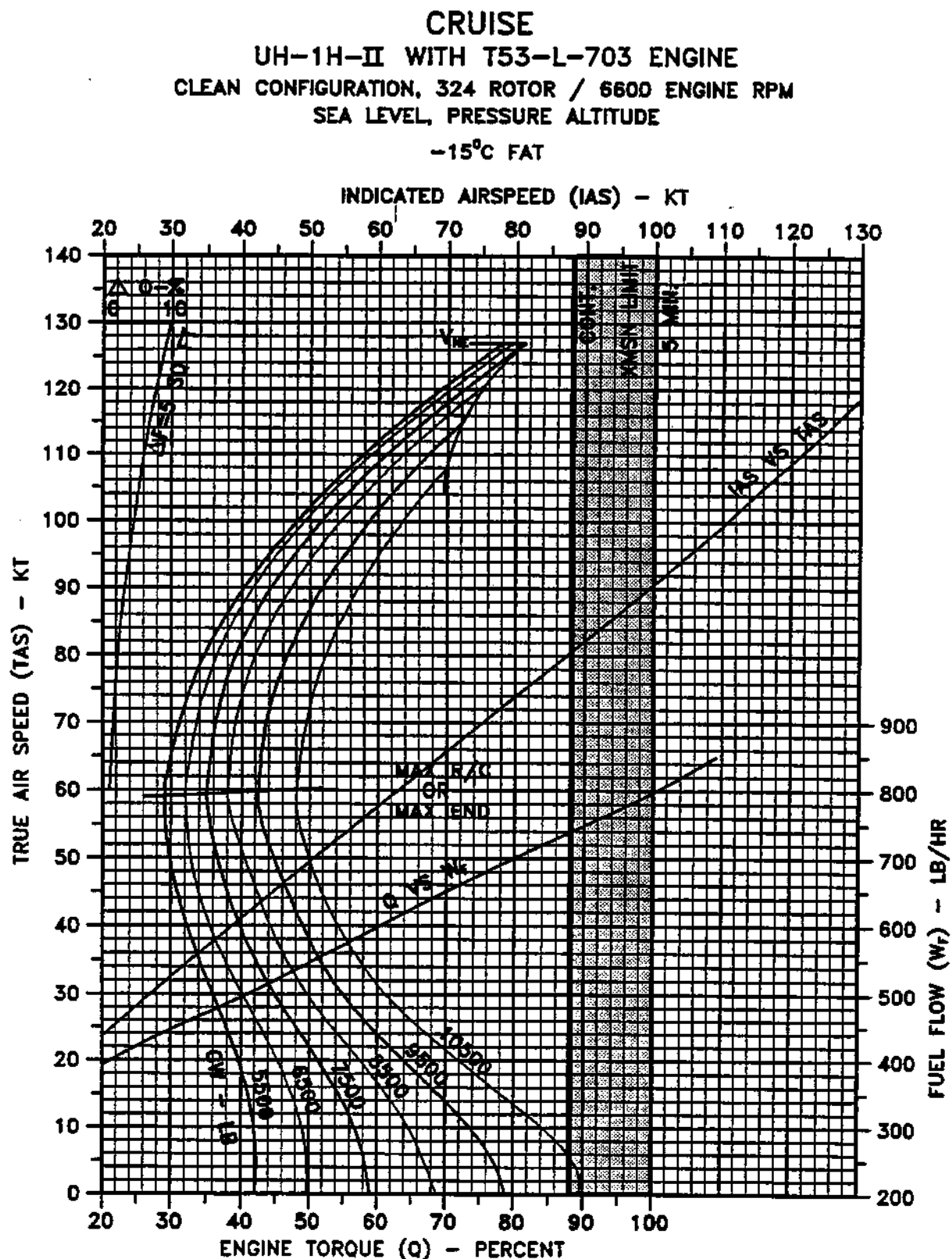
UH-1H-II-FM-7-15-10

Figure 7-13. Cruise chart, clean configuration, -30°C, 18000 feet, (Sheet 10 of 61)



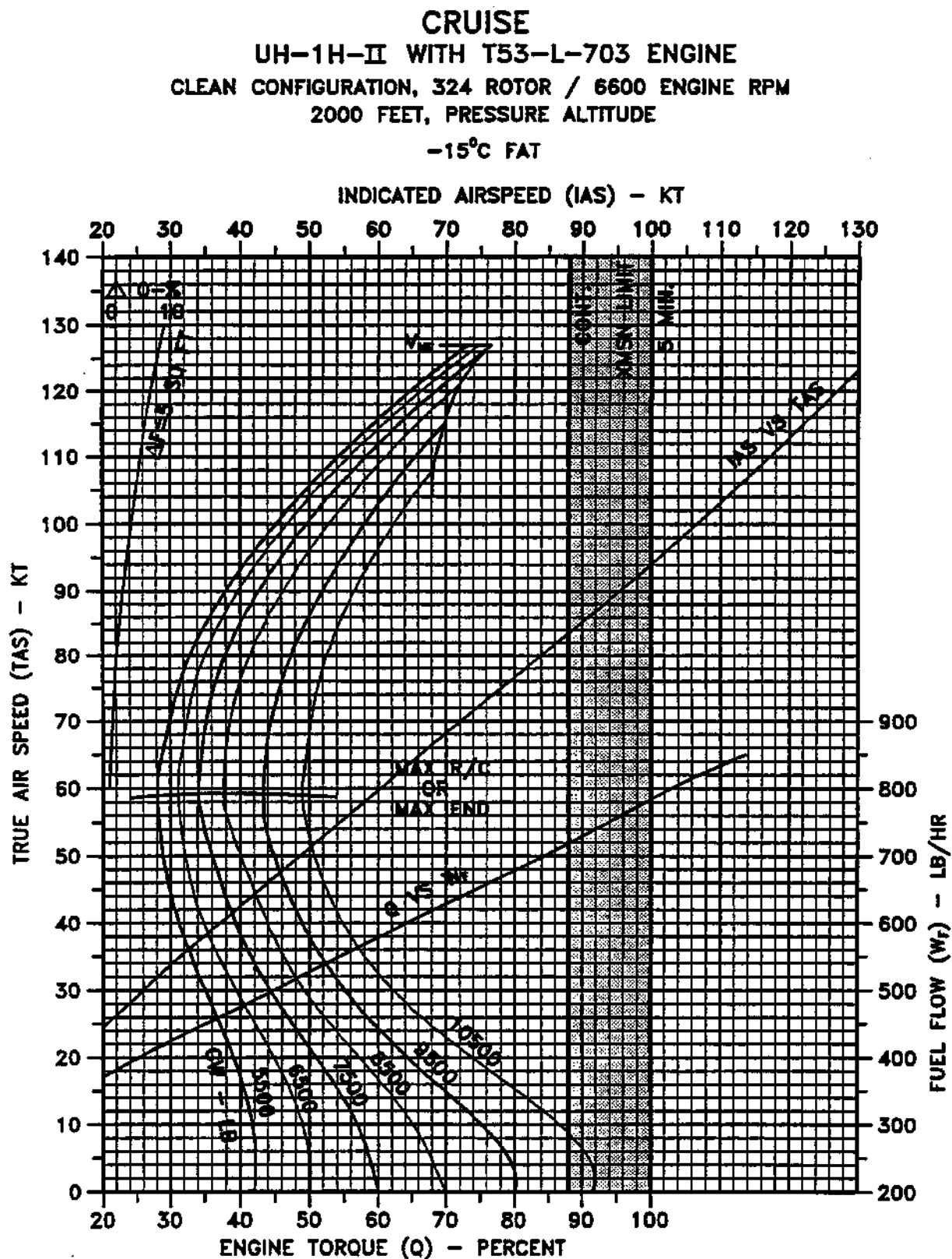
UH-1H-II-FM-7-15-11

Figure 7-13. Cruise chart, clean configuration, -30°C, 20000 feet, (Sheet 11 of 61)



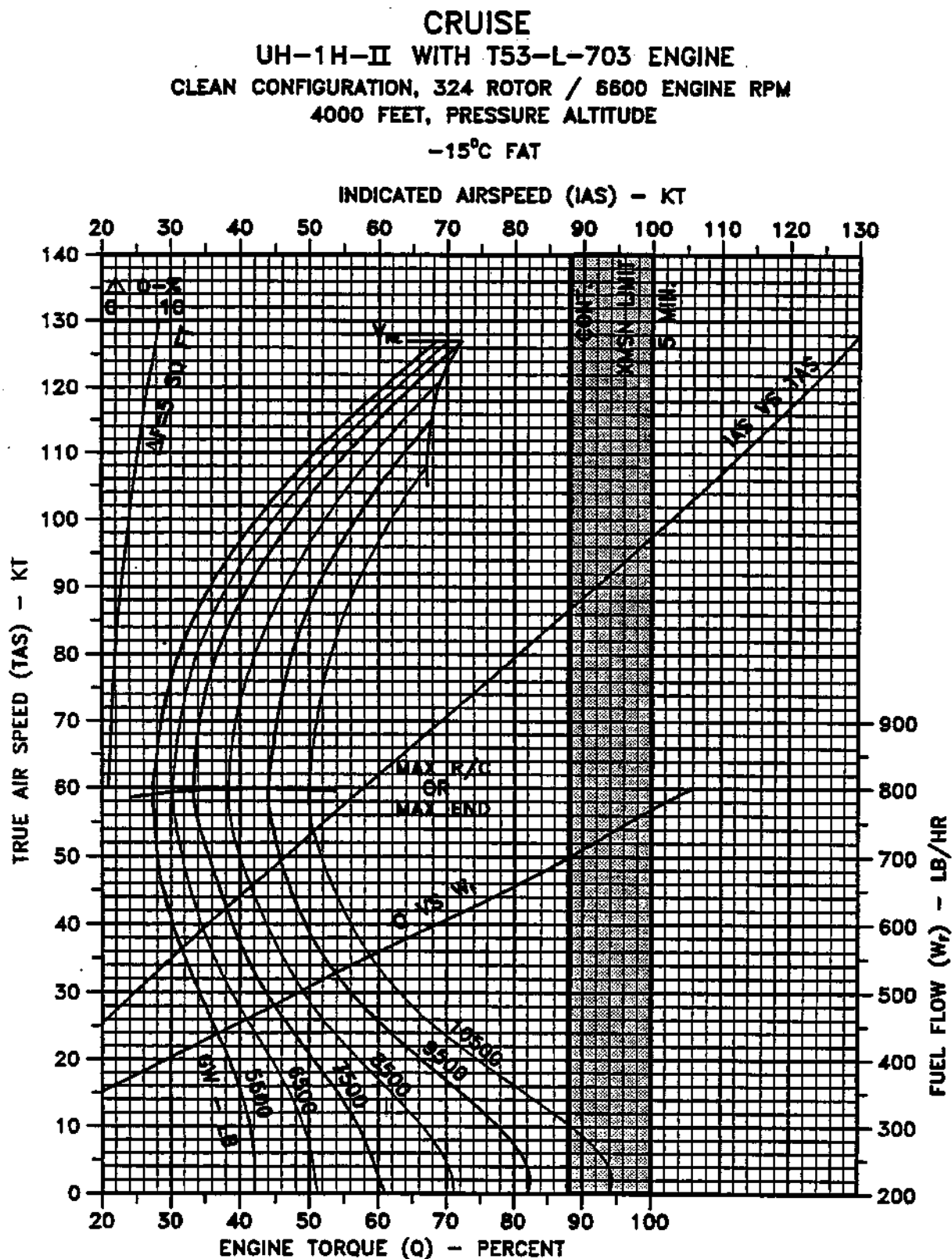
UH-1H-II-FM-7-15-12

Figure 7-13. Cruise chart, clean configuration, -15°C, sea level, (Sheet 12 of 61)



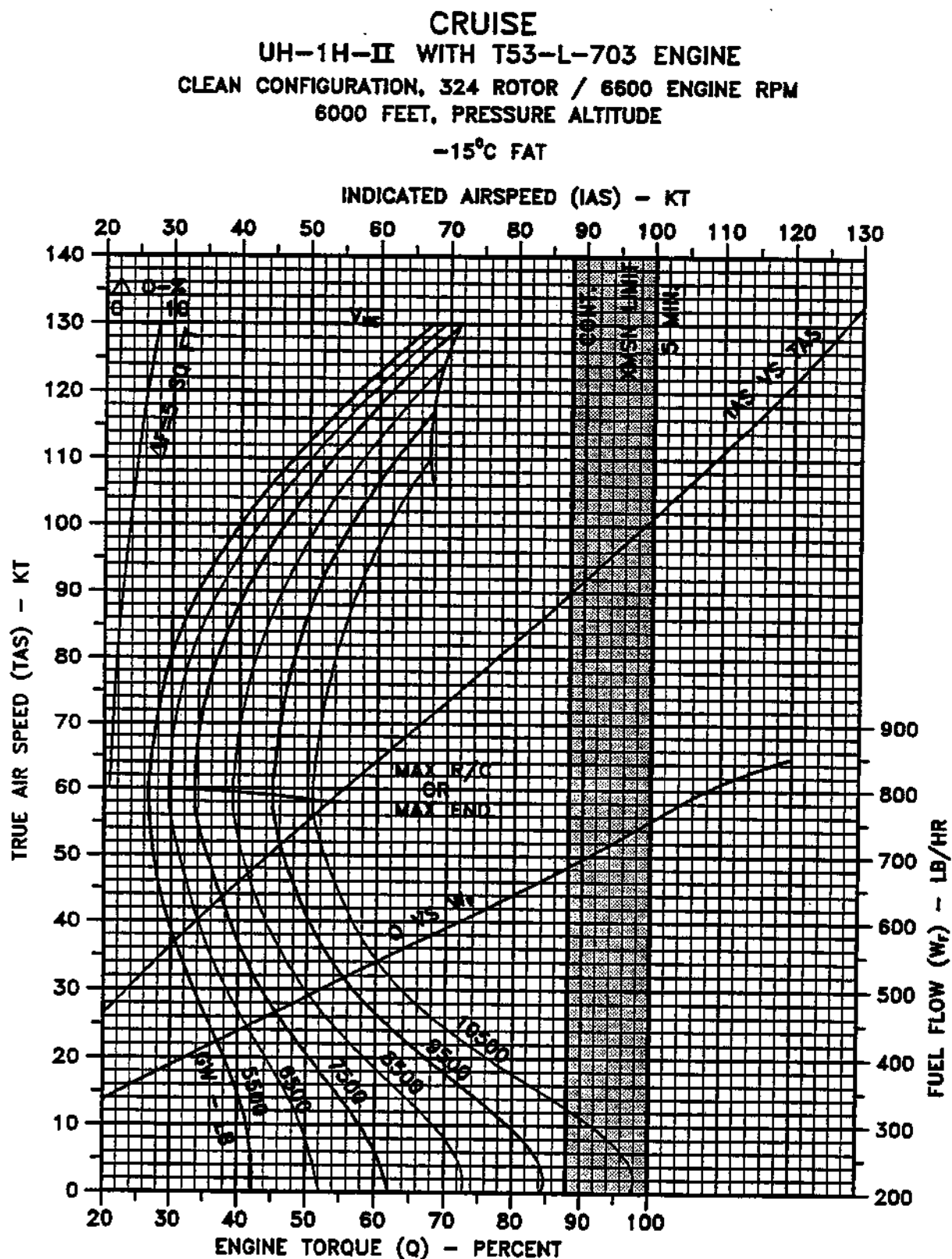
UH-1H-II-FM-7-15-13

Figure 7-13. Cruise chart, clean configuration, -15°C, 2000 feet, (Sheet 13 of 61)



UH-1H-II-FM-7-15-14

Figure 7-13. Cruise chart, clean configuration, -15°C, 4000 feet, (Sheet 14 of 61)

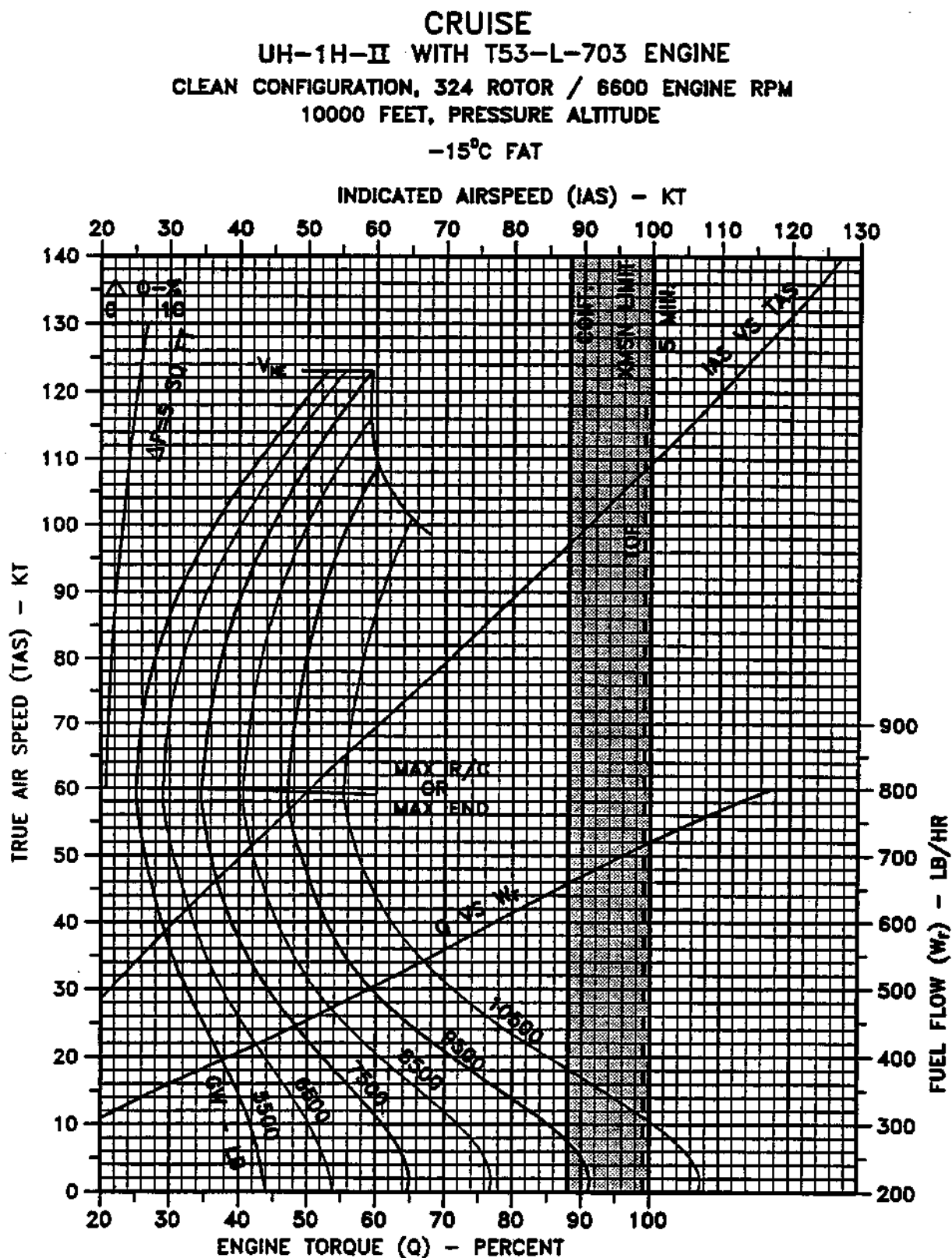


UH-1H-II-FM-7-15-15

Figure 7-13. Cruise chart, clean configuration, -15°C, 6000 feet, (Sheet 15 of 61)

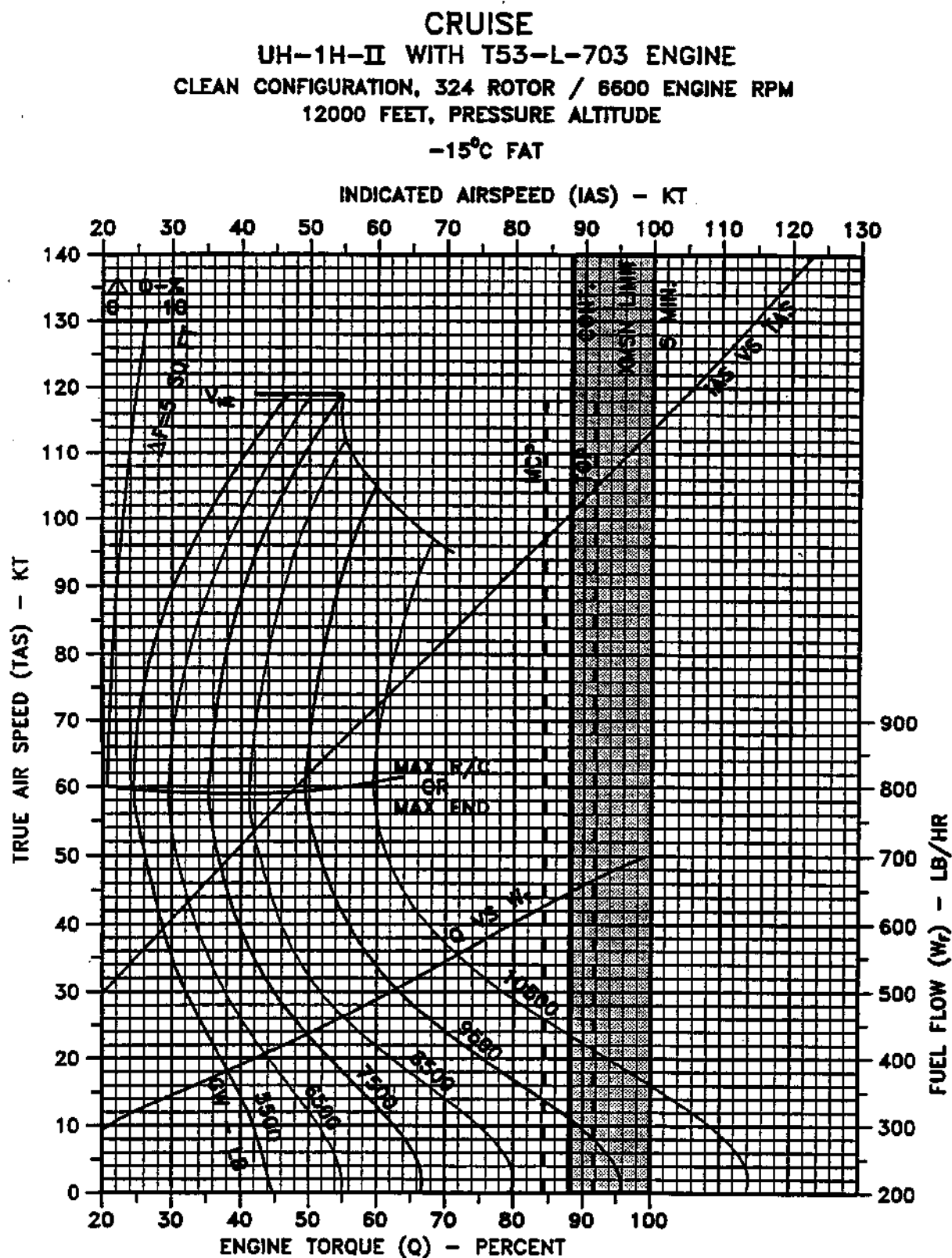


Figure 7-13. Cruise chart, clean configuration, -15°C, 8000 feet, (Sheet 16 of 61)



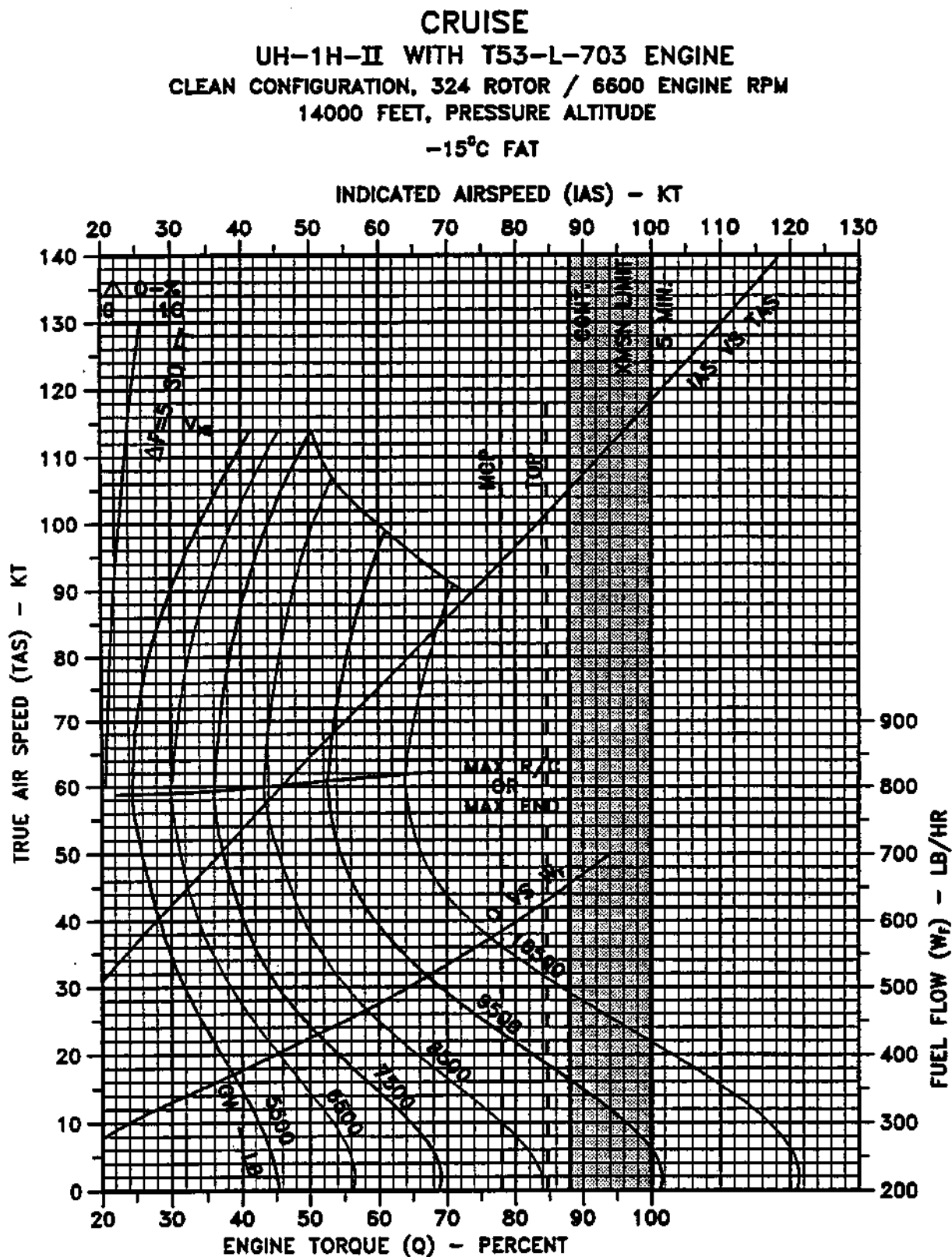
UH-1H-II-FM-7-15-17

Figure 7-13. Cruise chart, clean configuration, -15°C, 10000 feet, (Sheet 17 of 61)



UH-1H-II-FM-7-15-18

Figure 7-13. Cruise chart, clean configuration, -15°C, 12000 feet, (Sheet 18 of 61)



UH-1H-II-FM-7-15-19

Figure 7-13. Cruise chart, clean configuration, -15°C, 14000 feet, (Sheet 19 of 61)

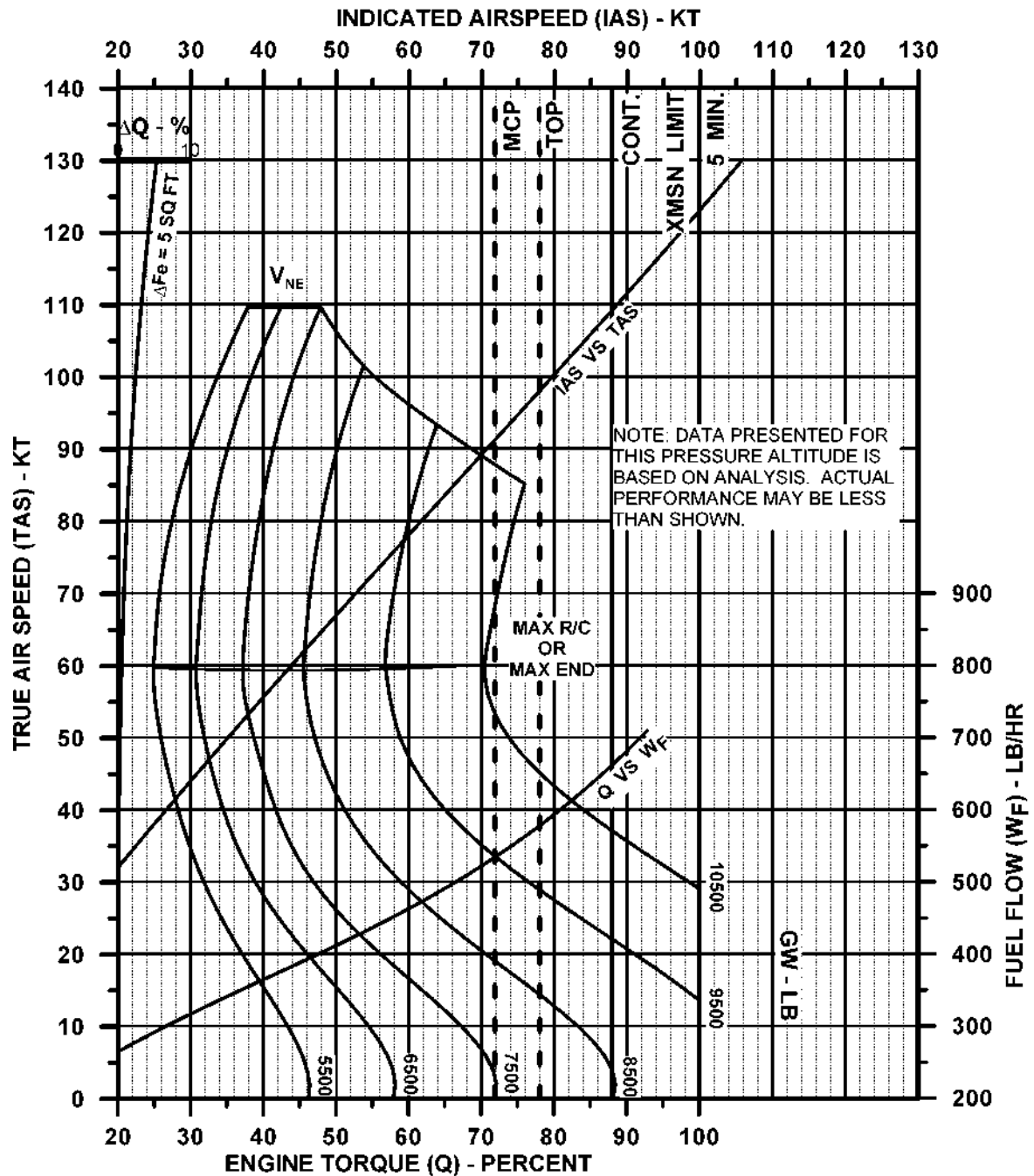
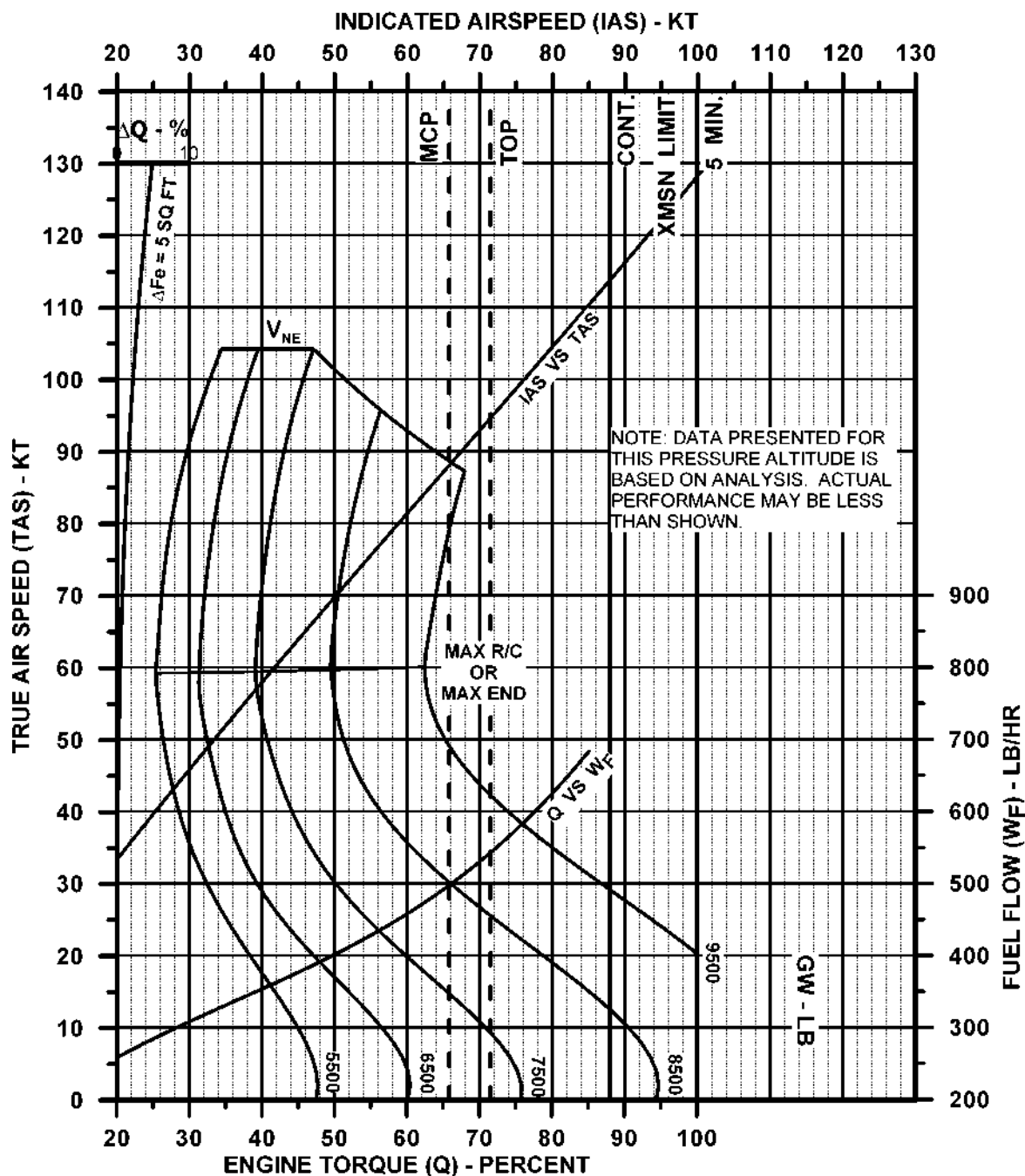


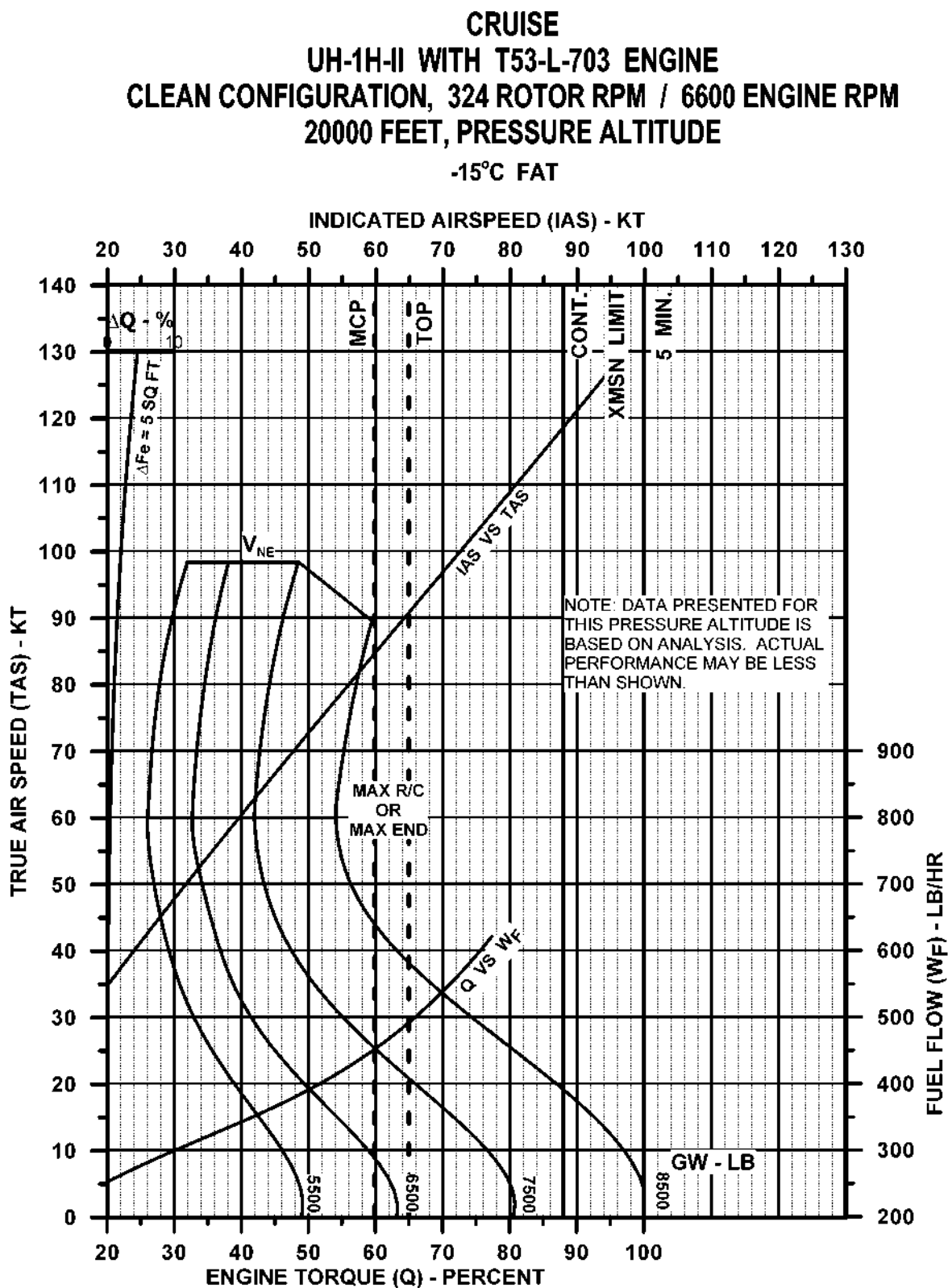
Figure 7-13. Cruise Chart, clean configuration, -15°C, 16000 feet, (Sheet 20 of 61)

CRUISE
UH-1H-II WITH T53-L-703 ENGINE
CLEAN CONFIGURATION, 324 ROTOR RPM / 6600 ENGINE RPM
18000 FEET, PRESSURE ALTITUDE
-15°C FAT



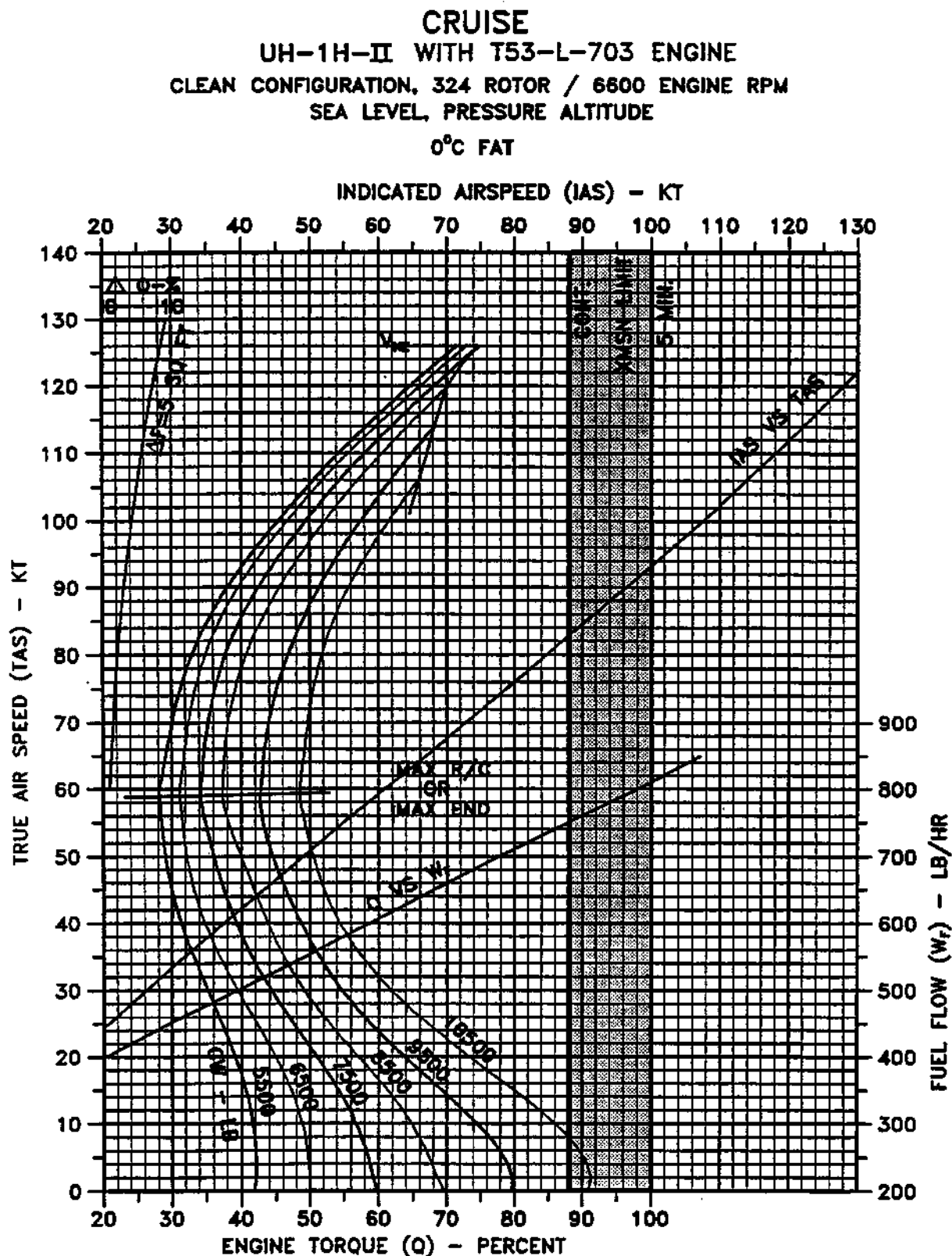
UH-1H-II-FM-7-15-21

Figure 7-13. Cruise Chart, clean configuration, -15°C, 18000 feet, (Sheet 21 of 61)



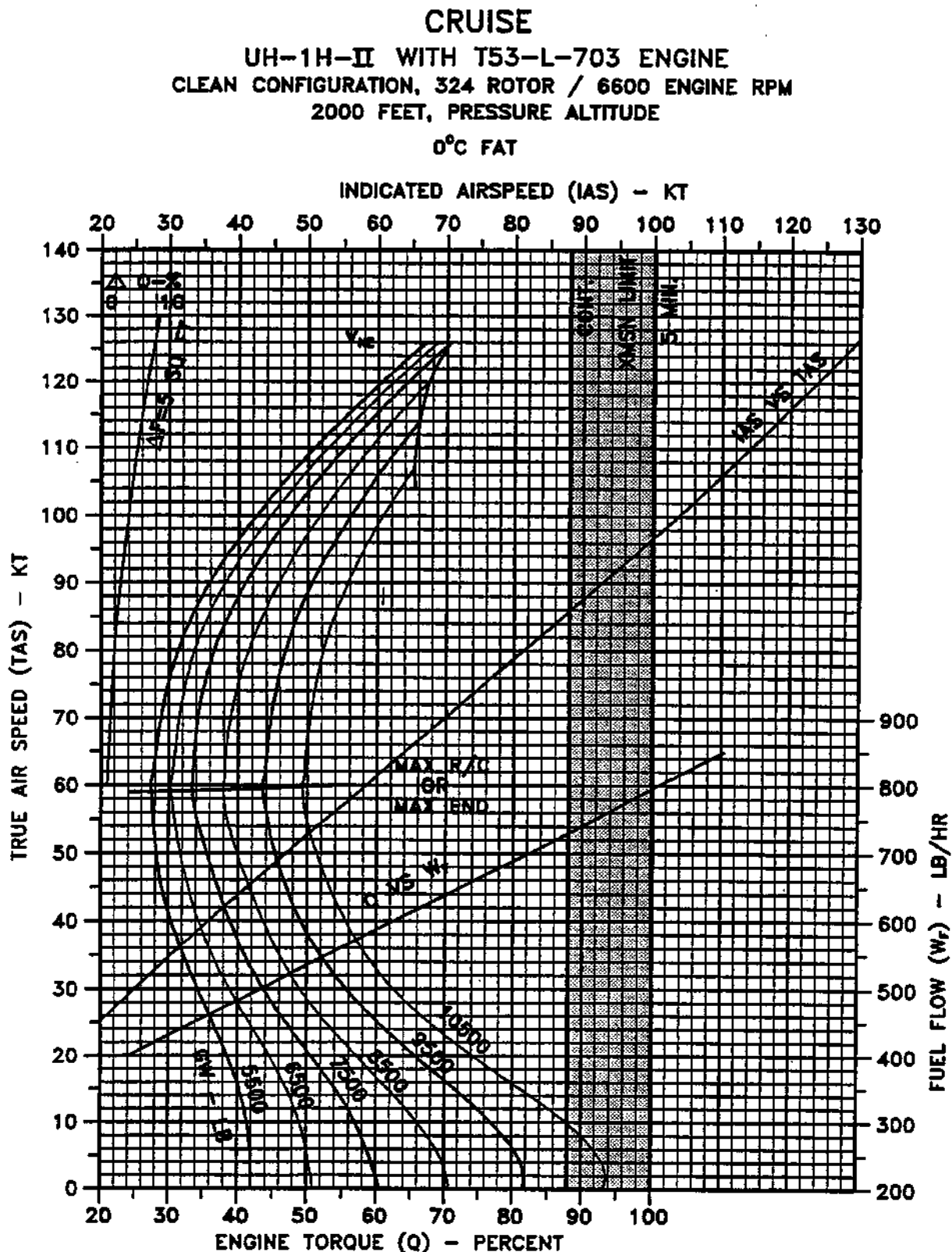
UH-1H-II-FM-7-15-22

Figure 7-13. Cruise Chart, clean configuration, -15°C, 20000 feet, (Sheet 22 of 61)



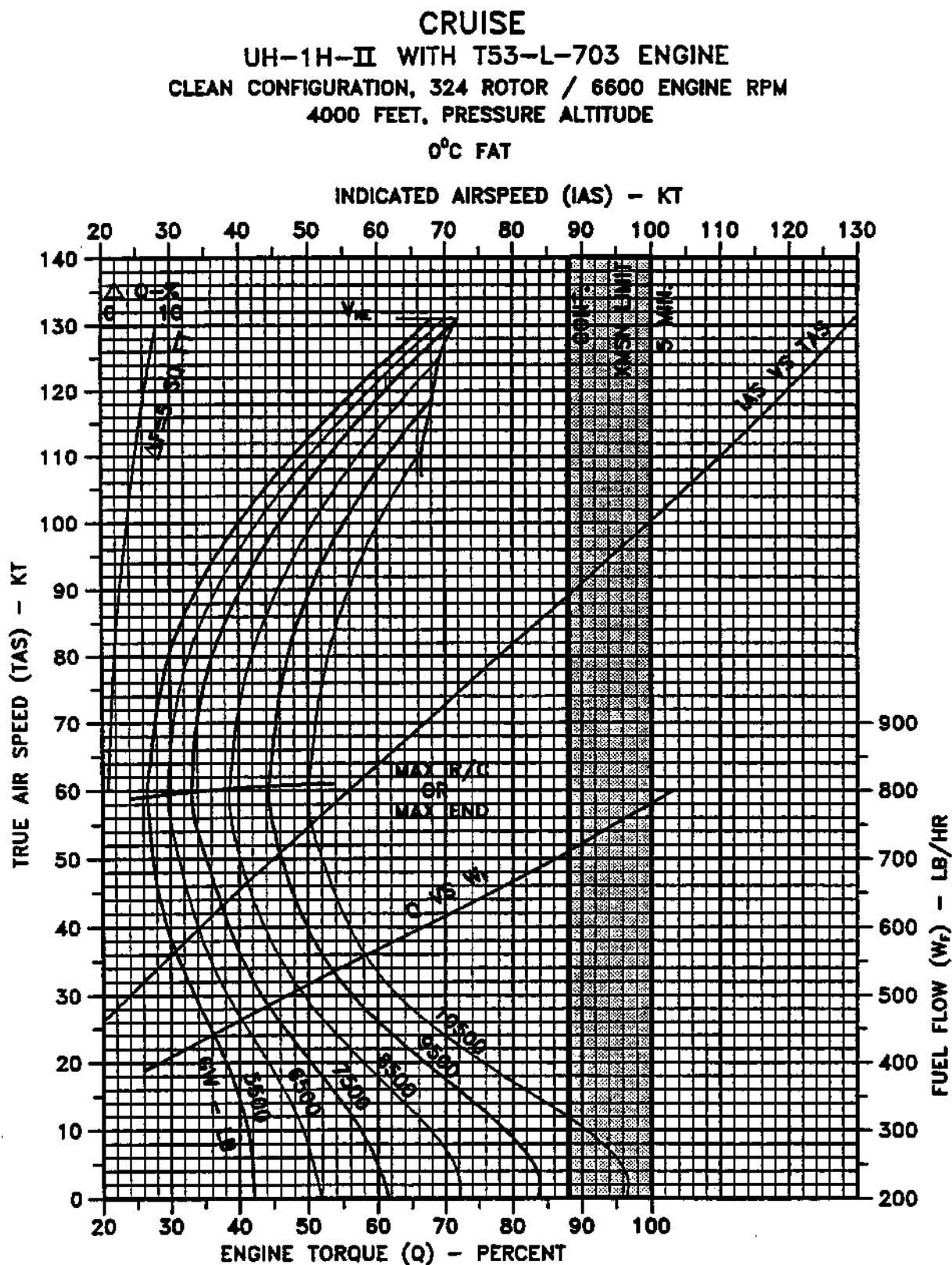
UH-1H-II-FM-7-15-23

Figure 7-13. Cruise chart, clean configuration, -0°C, sea level, (Sheet 23 of 61)



UH-1H-II-FM-7-15-24

Figure 7-13. Cruise chart, clean configuration, -0°C, 2000 feet, (Sheet 24 of 61)



UH-1H-II-FM-7-15-25

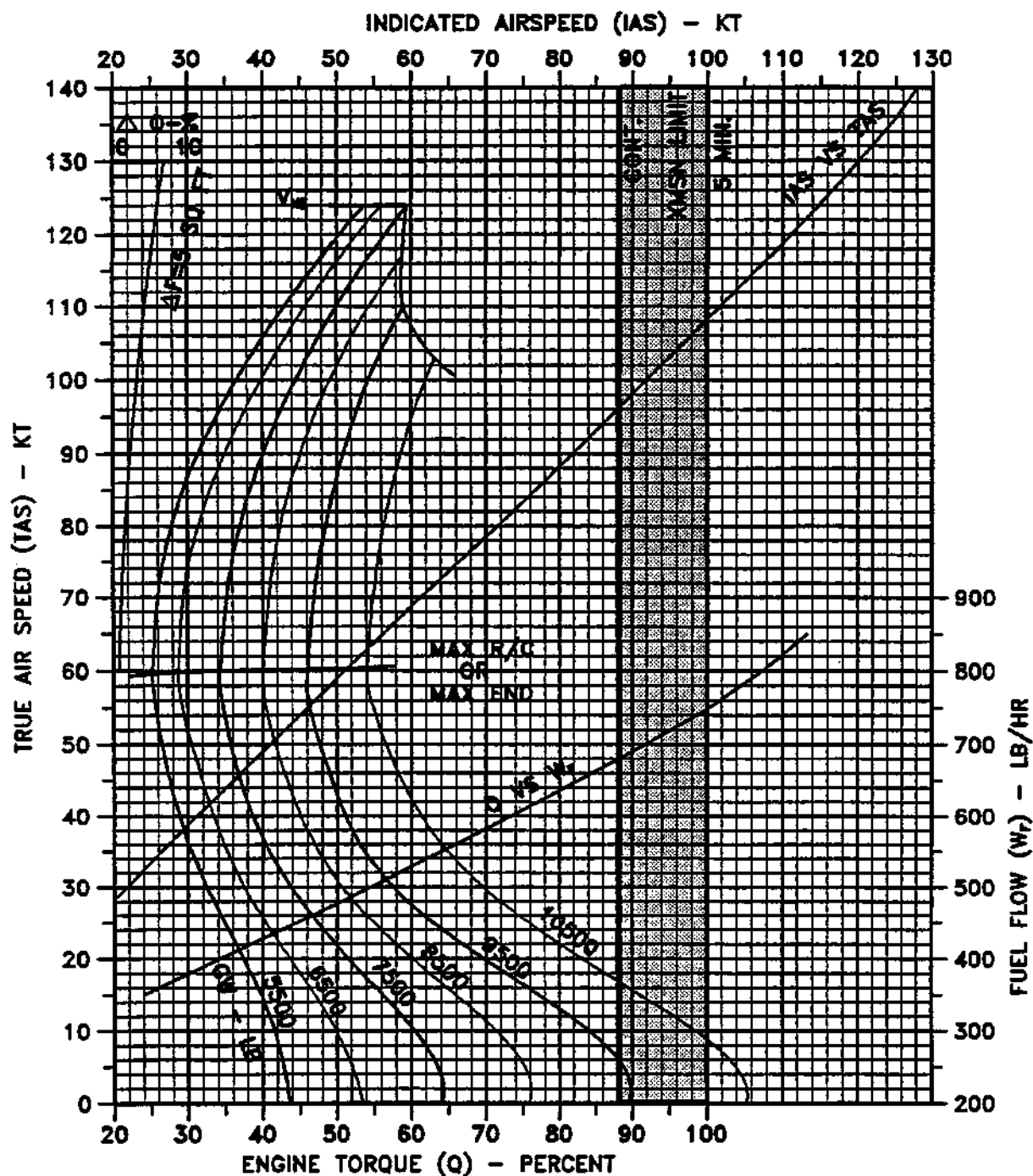
Figure 7-13. Cruise chart, clean configuration, -0°C, 4000 feet, (Sheet 25 of 61)



Figure 7-13. Cruise chart, clean configuration, -0°C , 6000 feet, (Sheet 26 of 61)

CRUISE

UH-1H-II WITH T53-L-703 ENGINE
 CLEAN CONFIGURATION, 324 ROTOR / 6600 ENGINE RPM
 8000 FEET, PRESSURE ALTITUDE
 0°C FAT



UH-1H-II-FM-7-15-27

Figure 7-13. Cruise chart, clean configuration, -0°C, 8000 feet, (Sheet 27 of 61)

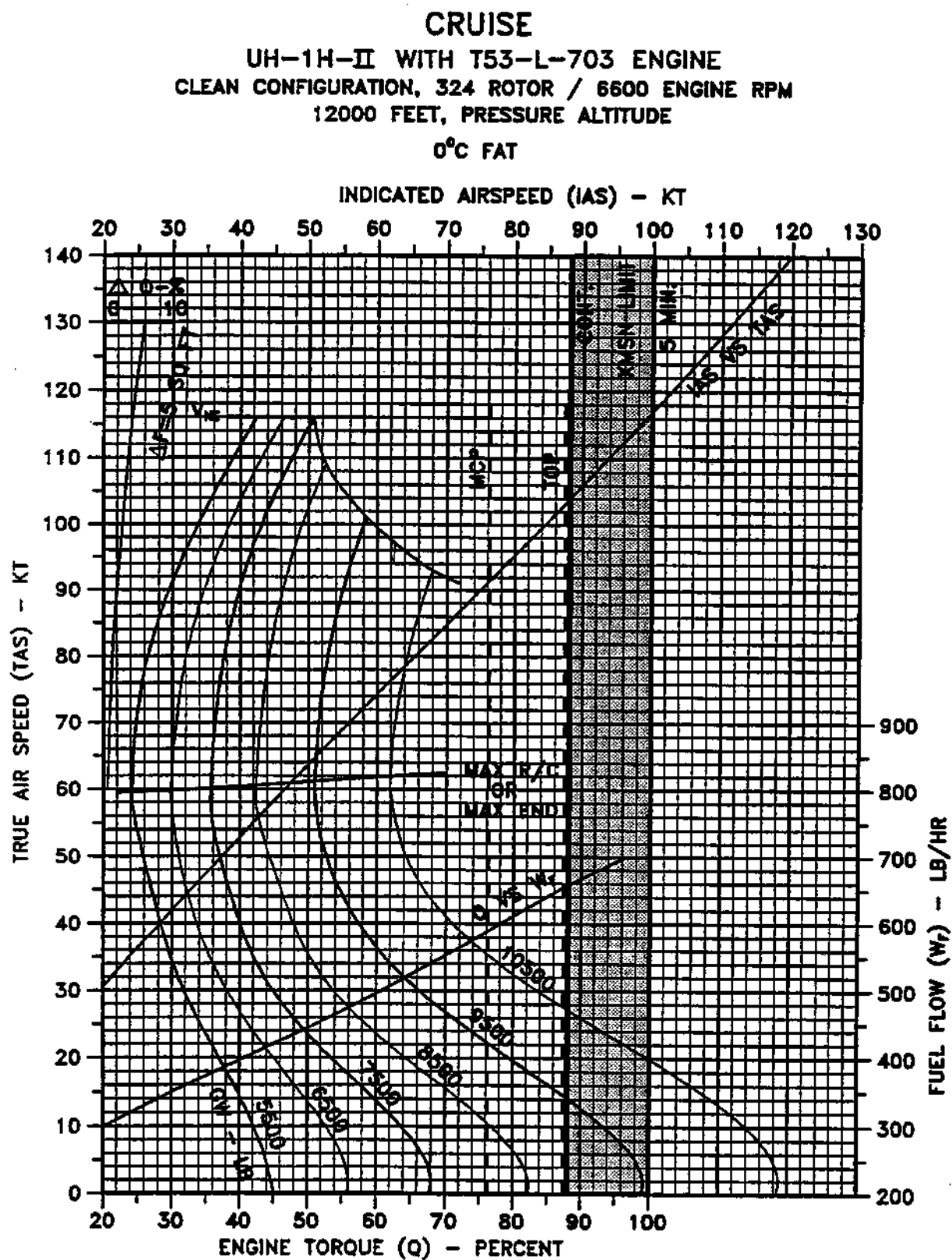
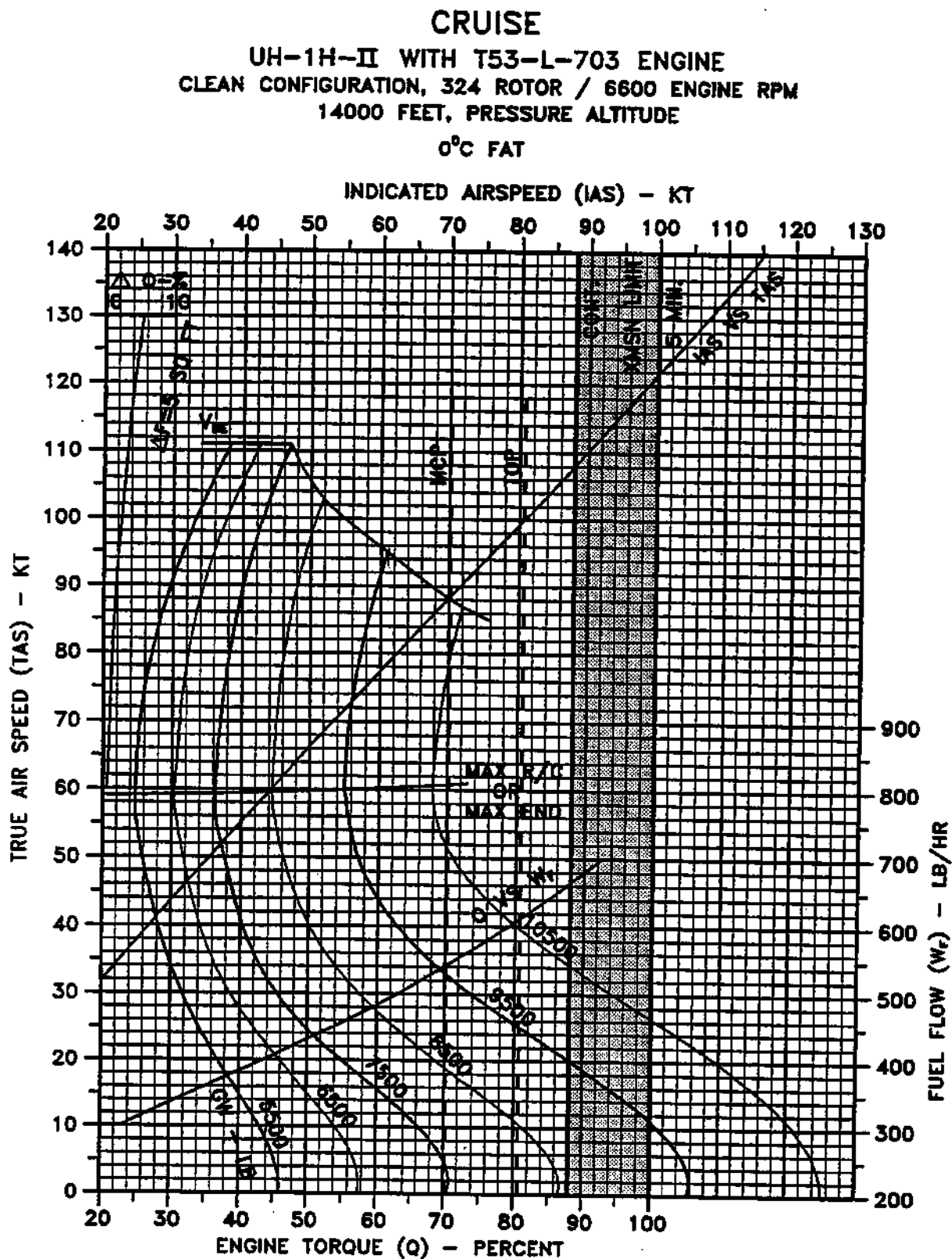
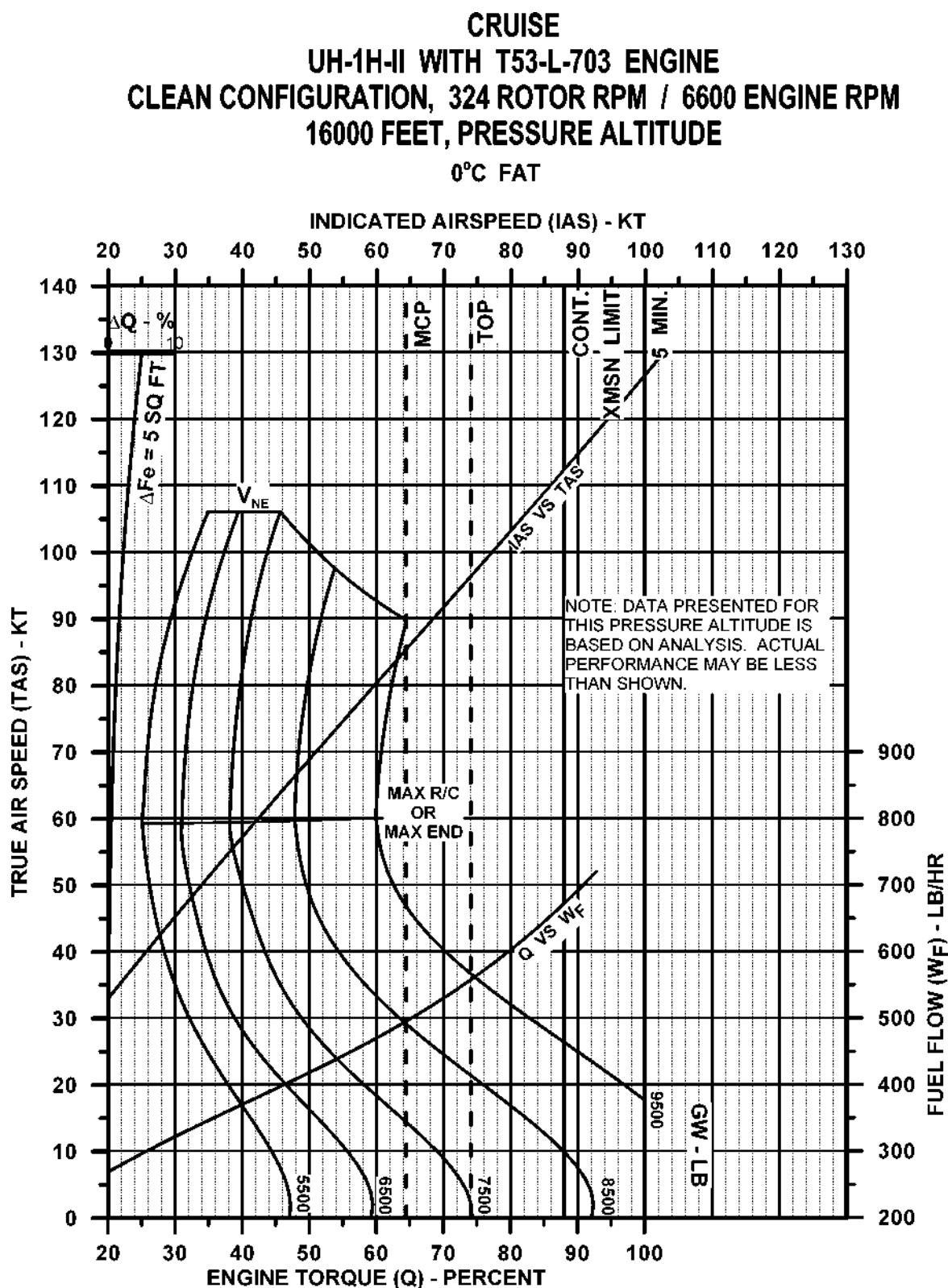


Figure 7-13. Cruise chart, clean configuration, -0°C, 12000 feet, (Sheet 29 of 61)



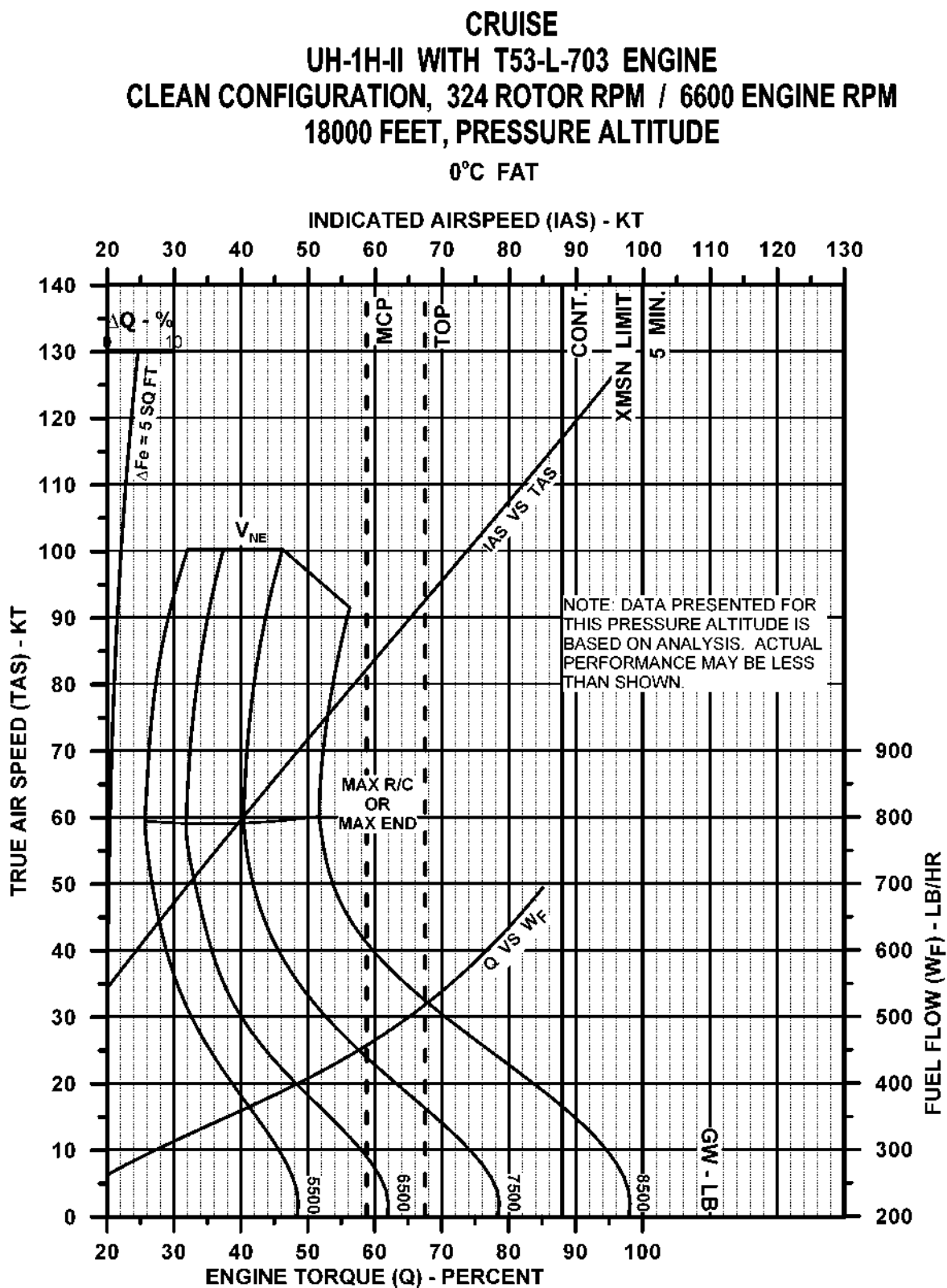
UH-1H-II-FM-7-15-30

Figure 7-13. Cruise chart, clean configuration, -0°C, 14000 feet, (Sheet 30 of 61)



UH-1H-II-FM-7-15-31

Figure 7-13. Cruise chart, clean configuration, -0°C, 16000 feet, (Sheet 31 of 61)

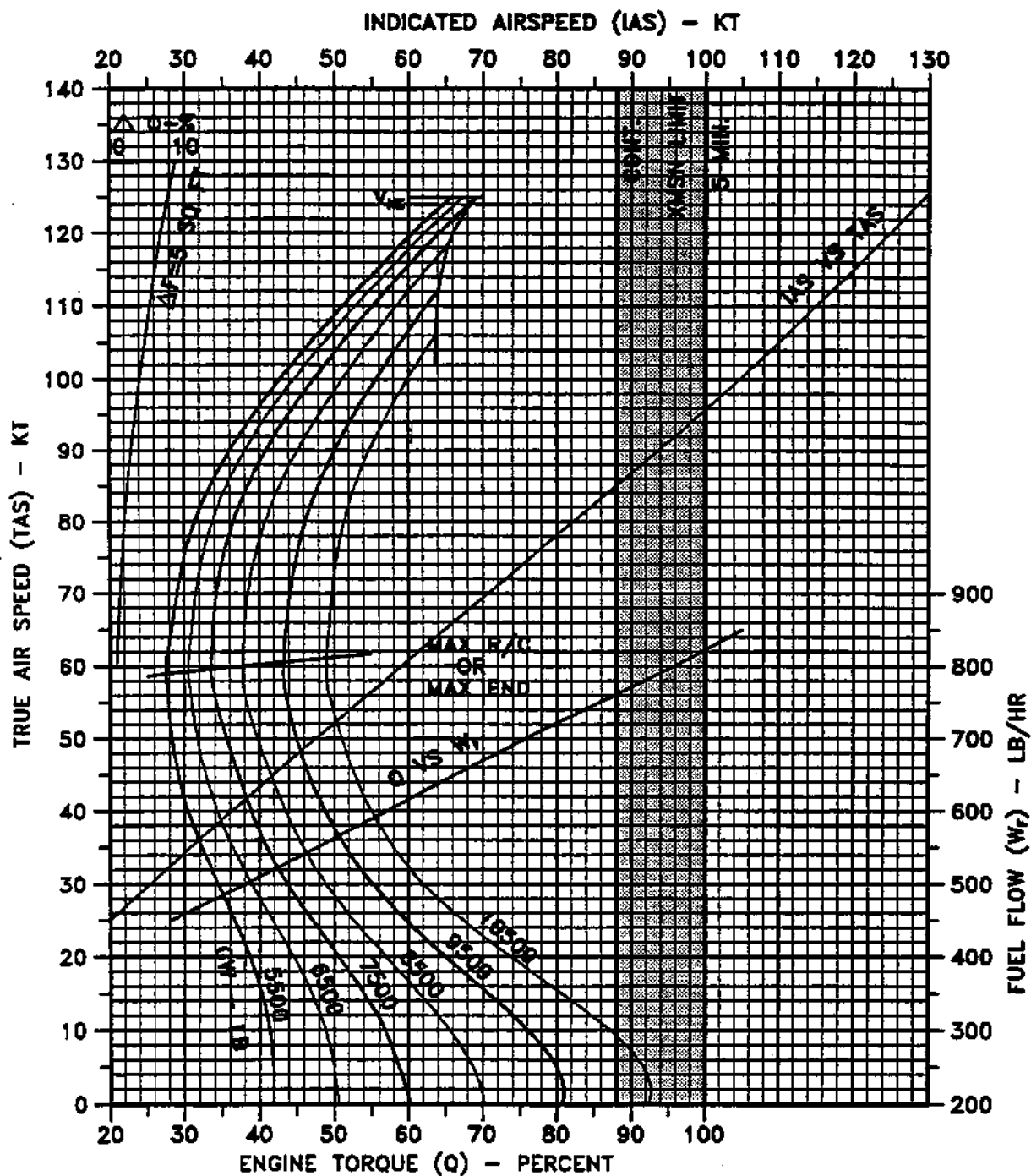


UH-1H-II-FM-7-15-32

Figure 7-13. Cruise chart, clean configuration, -0°C, 18000 feet, (Sheet 32 of 61)

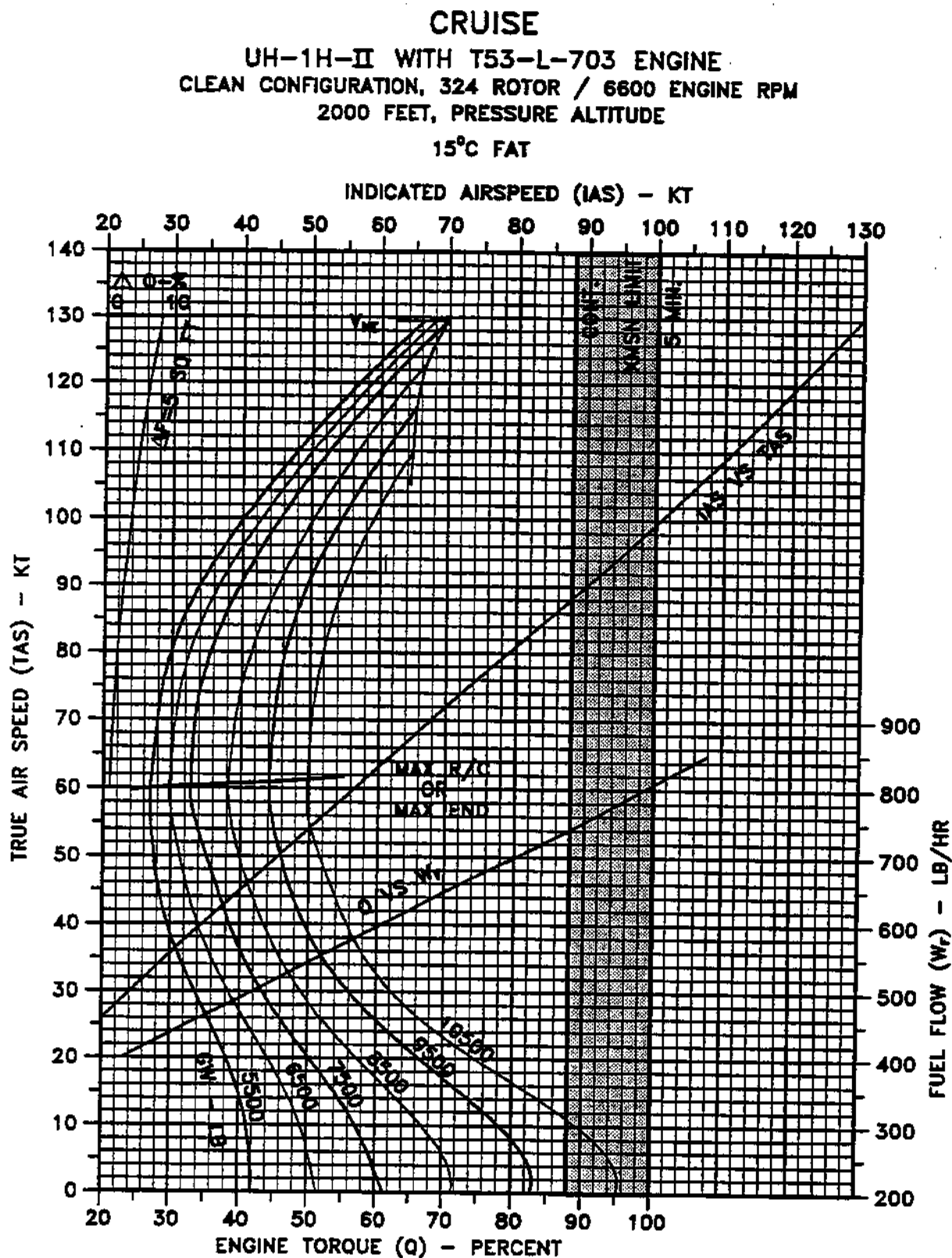
CRUISE

UH-1H-II WITH T53-L-703 ENGINE
 CLEAN CONFIGURATION, 324 ROTOR / 6600 ENGINE RPM
 SEA LEVEL, PRESSURE ALTITUDE
 15°C FAT



UH-1H-II-FM-7-15-33

Figure 7-13. Cruise chart, clean configuration, +15°C, sea level, (Sheet 33 of 61)

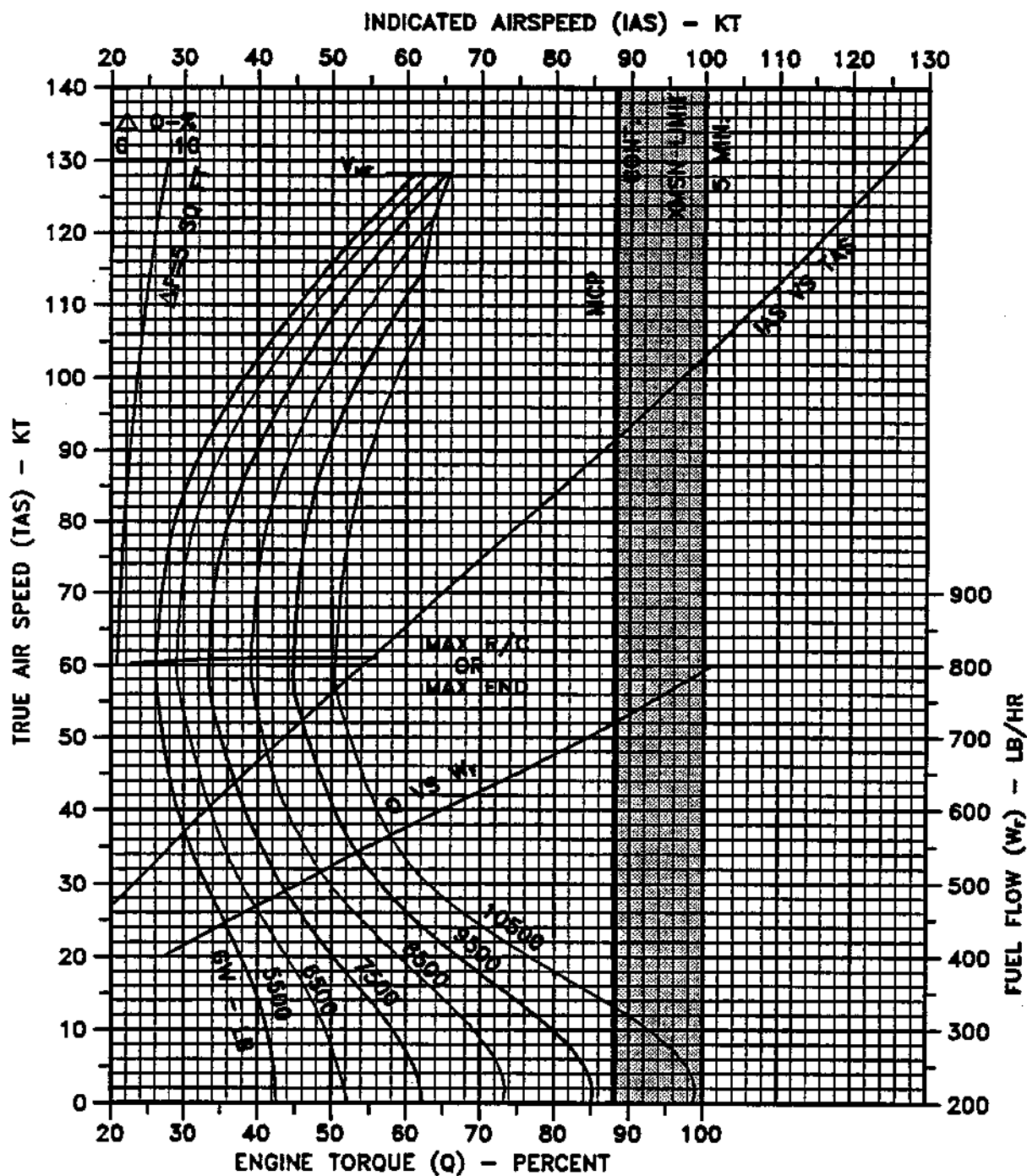


UH-1H-II-FM-7-15-34

Figure 7-13. Cruise chart, clean configuration, +15°C, 2000 feet, (Sheet 34 of 61)

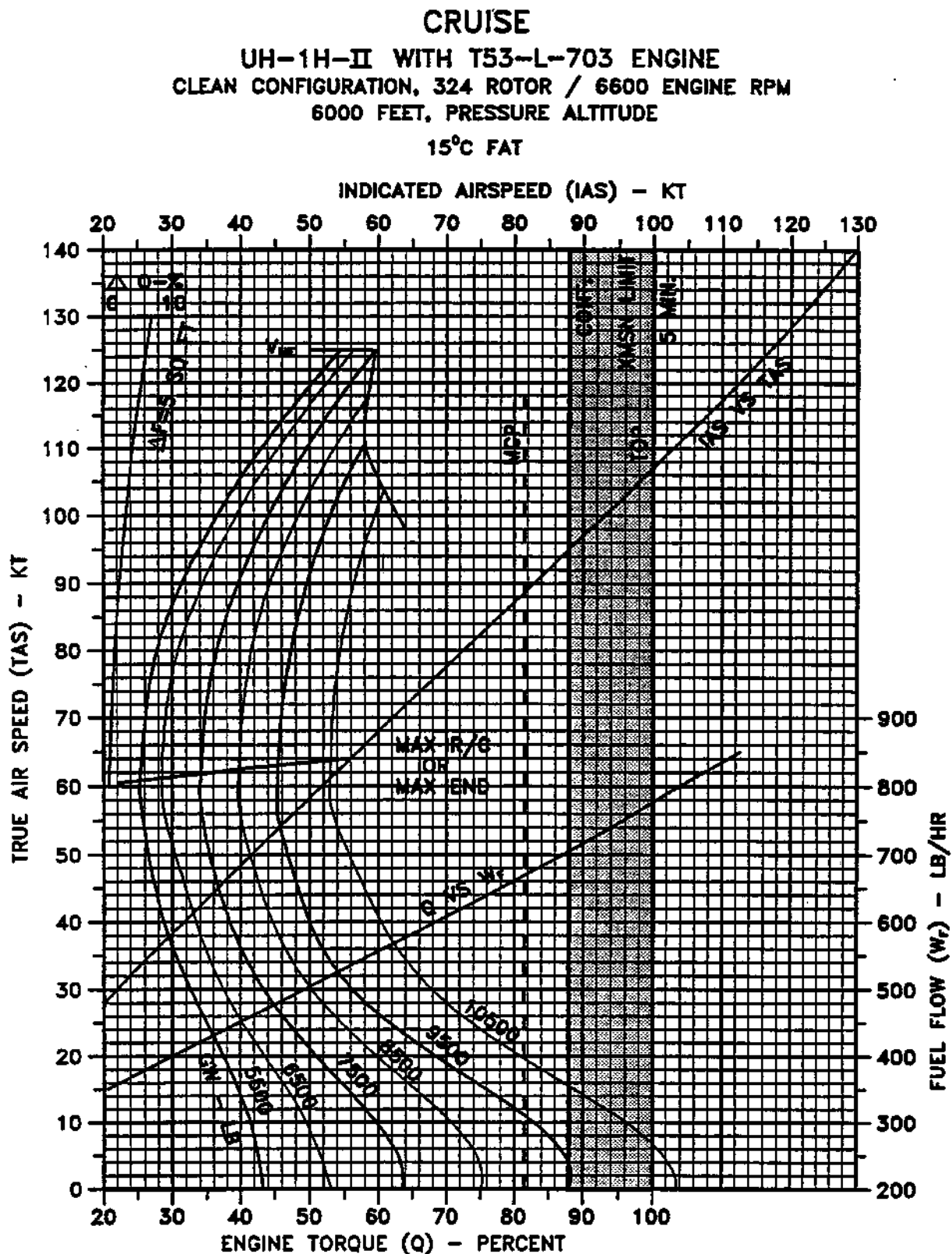
CRUISE

UH-1H-II WITH T53-L-703 ENGINE
 CLEAN CONFIGURATION, 324 ROTOR / 8600 ENGINE RPM
 4000 FEET, PRESSURE ALTITUDE
 15°C FAT



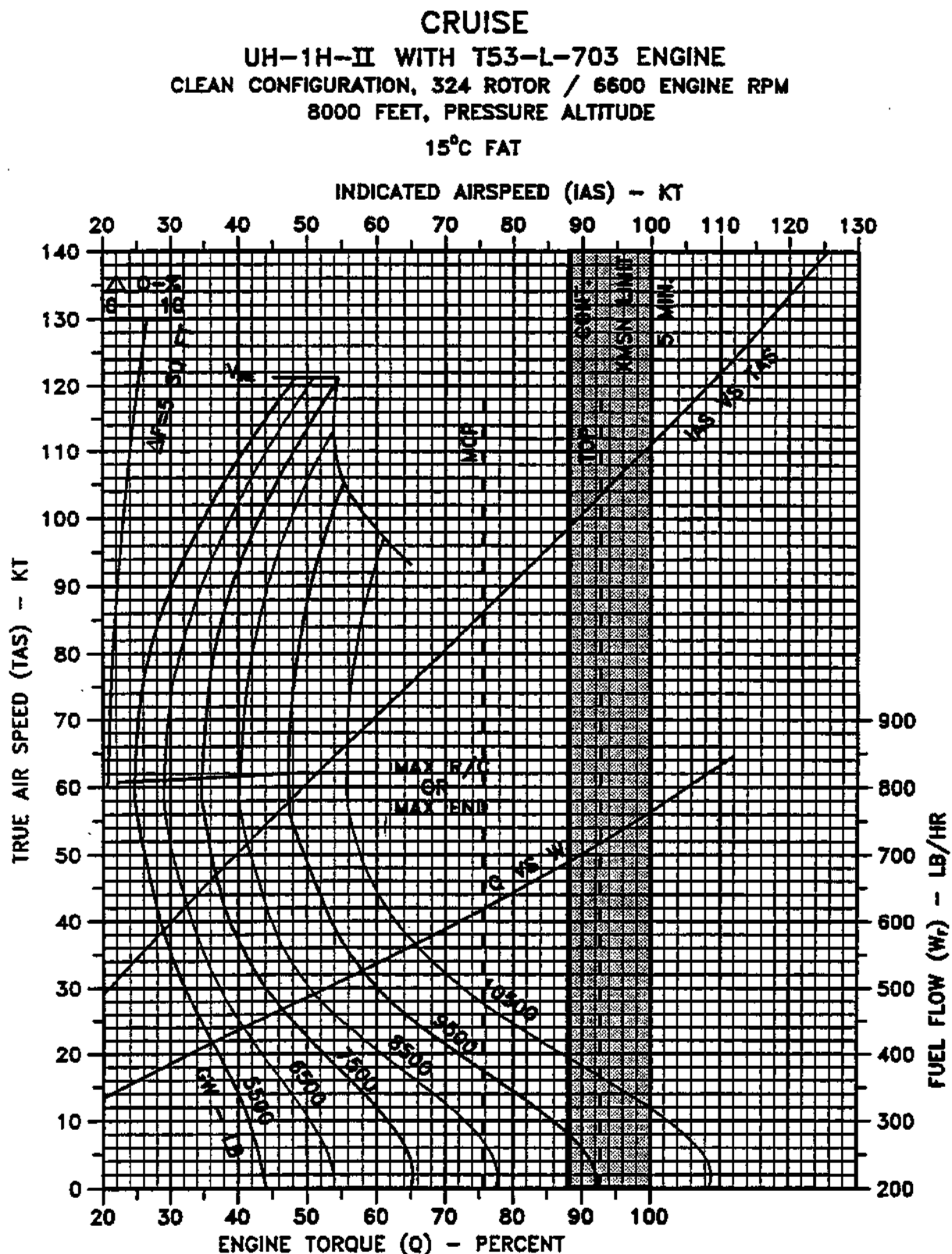
UH-1H-II-FM-7-15-35

Figure 7-13. Cruise chart, clean configuration, +15°C, 4000 feet, (Sheet 35 of 61)



UH-1H-II-FM-7-15-36

Figure 7-13. Cruise chart, clean configuration, +15°C, 6000 feet, (Sheet 36 of 61)

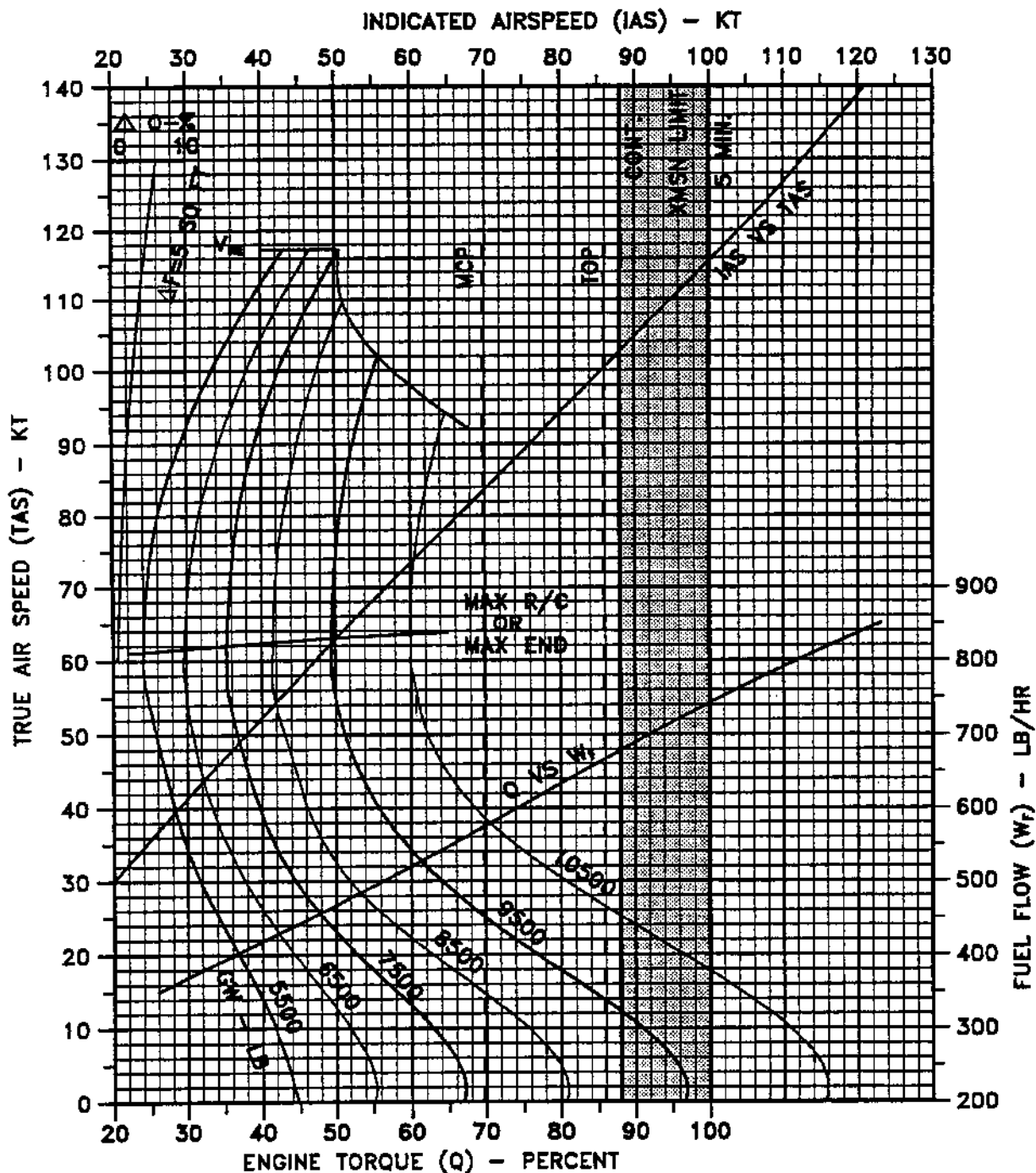


UH-1H-II-FM-7-15-37

Figure 7-13. Cruise chart, clean configuration, +15°C, 8000 feet, (Sheet 37 of 61)

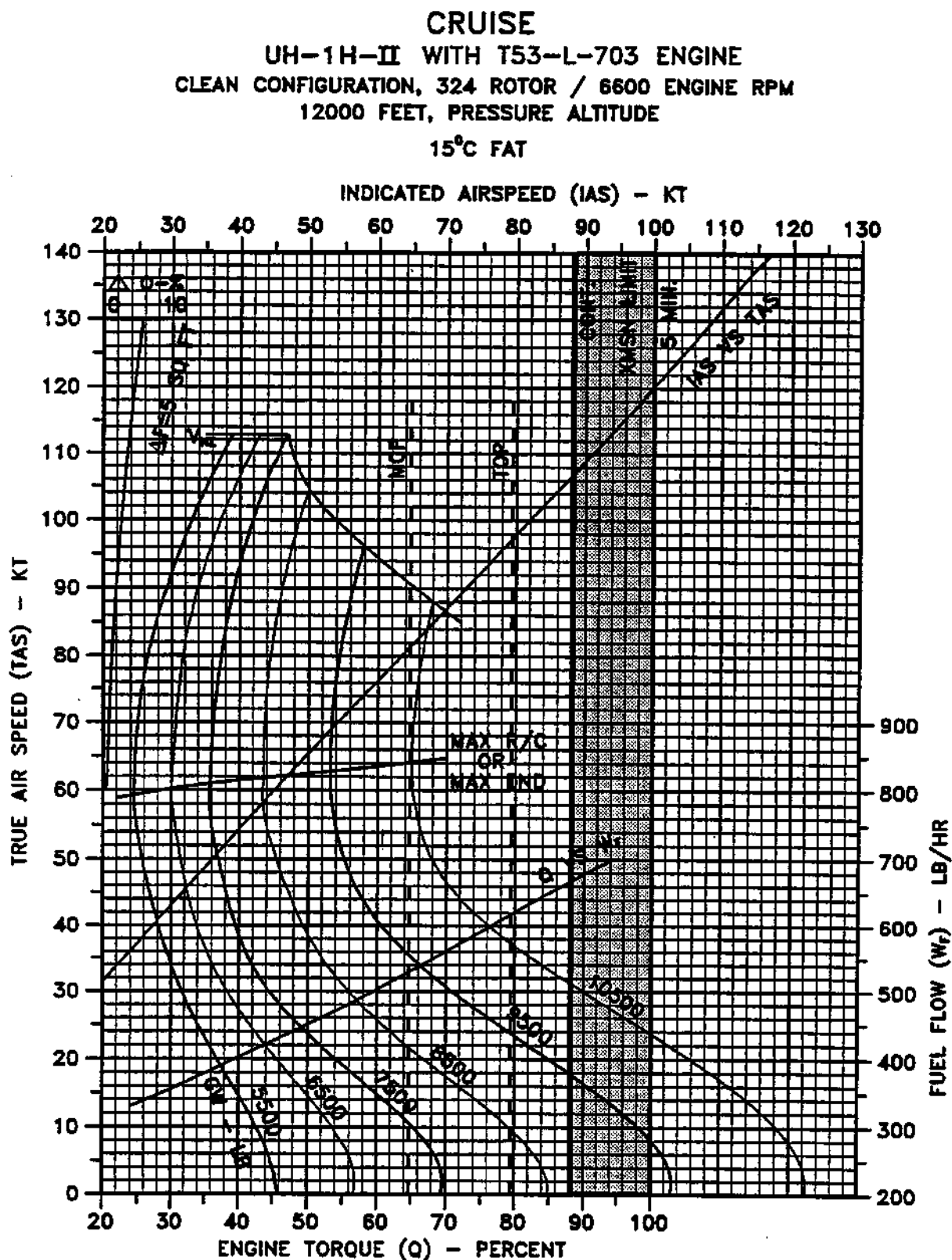
CRUISE

UH-1H-II WITH T53-L-703 ENGINE
CLEAN CONFIGURATION, 324 ROTOR / 6600 ENGINE RPM
10000 FEET, PRESSURE ALTITUDE
15°C FAT



UH-1H-II-FM-7-15-38

Figure 7-13. Cruise chart, clean configuration, +15°C, 10000 feet, (Sheet 38 of 61)



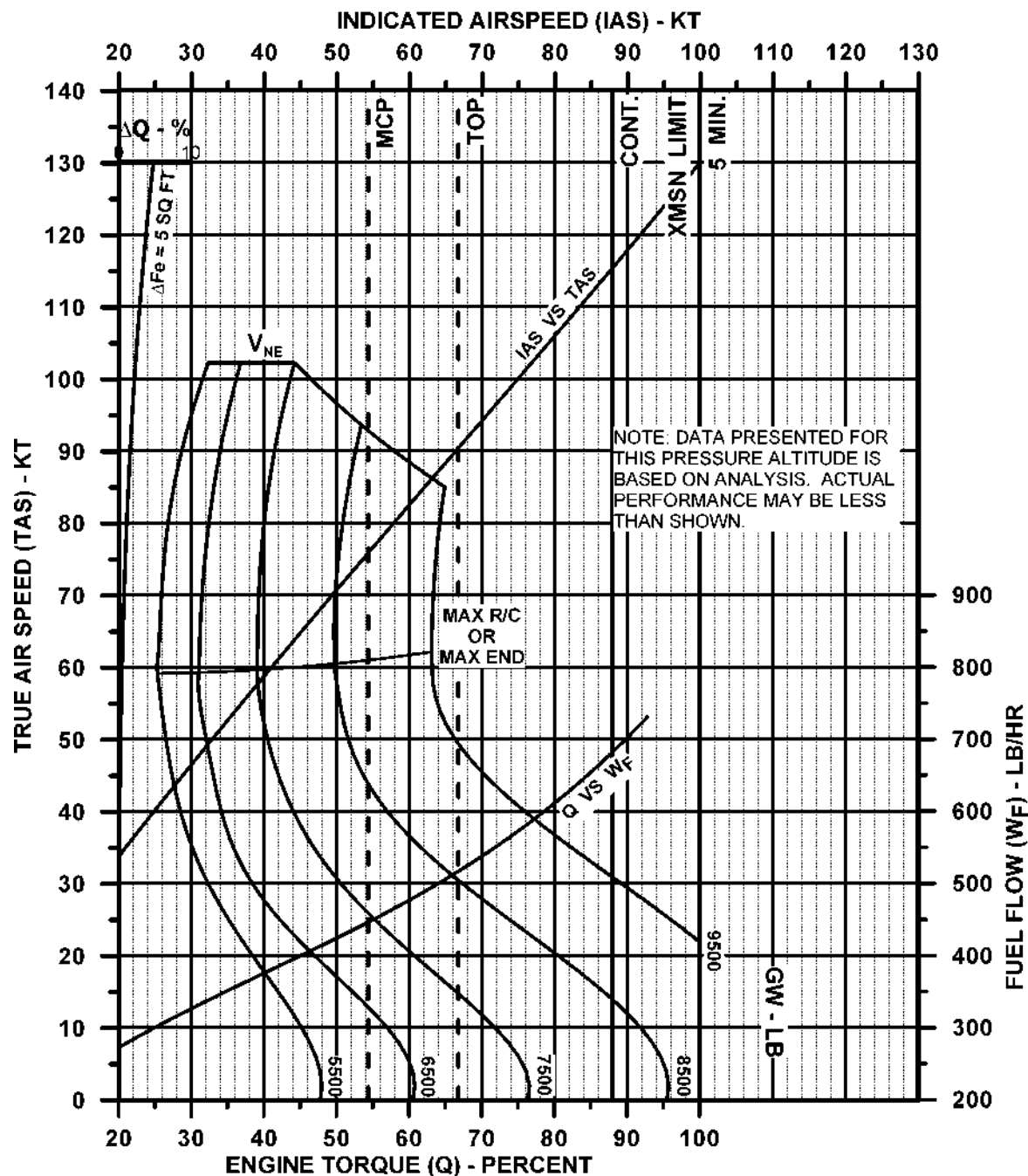
UH-1H-II-FM-7-15-39

Figure 7-13. Cruise chart, clean configuration, +15°C, 12000 feet, (Sheet 39 of 61)



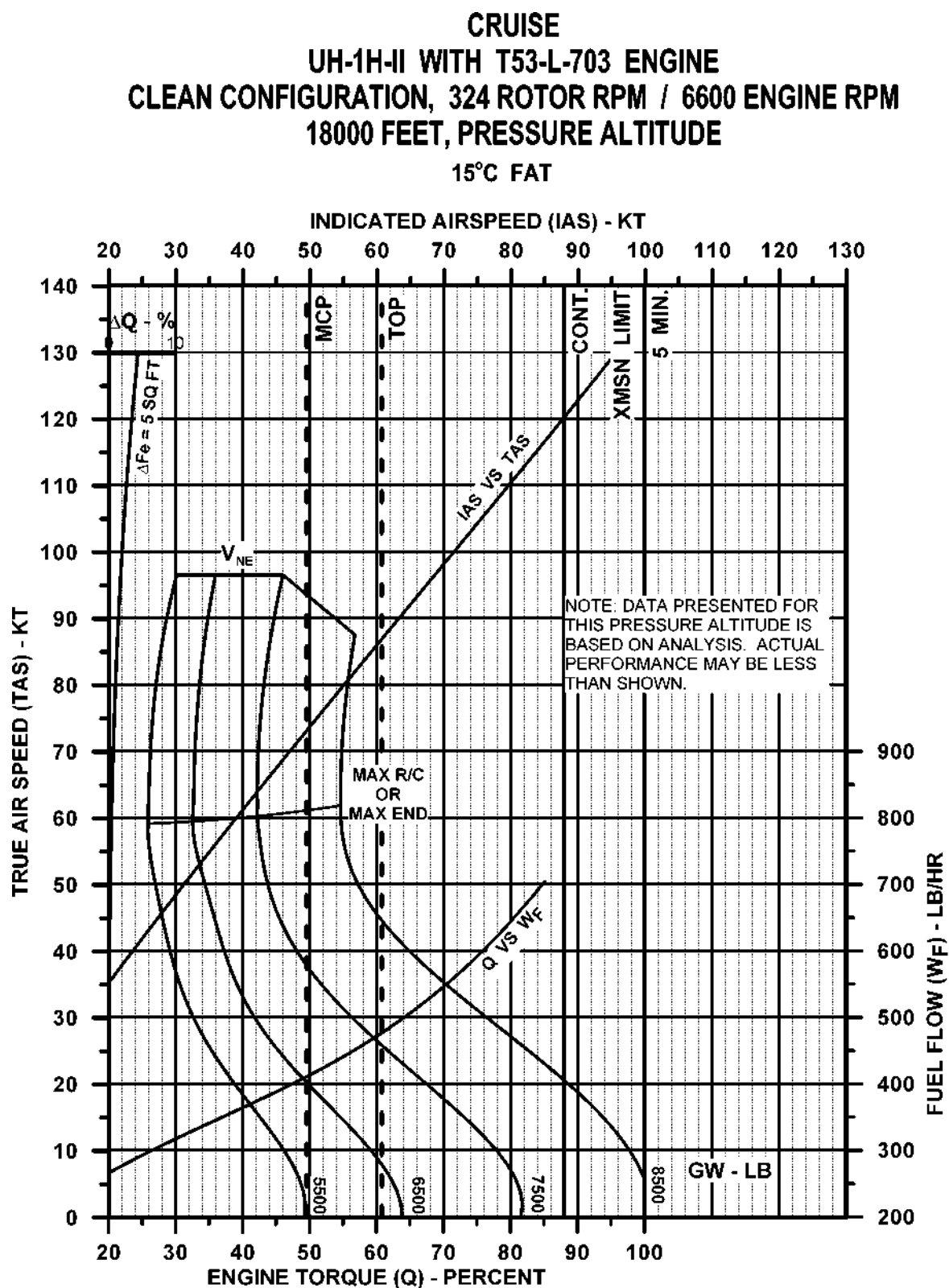
7-00-00
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CRUISE
UH-1H-II WITH T53-L-703 ENGINE
CLEAN CONFIGURATION, 324 ROTOR RPM / 6600 ENGINE RPM
16000 FEET, PRESSURE ALTITUDE
15°C FAT



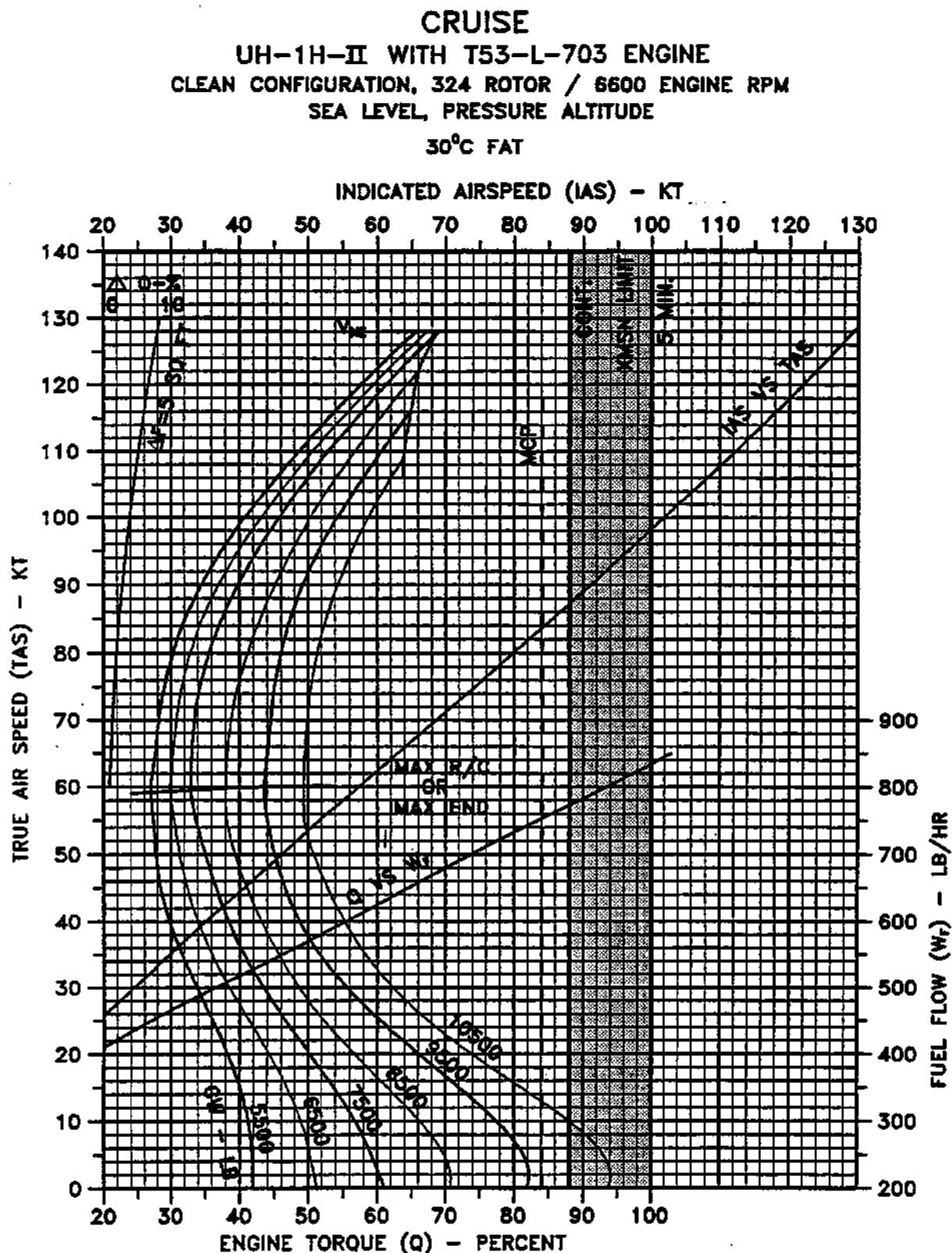
UH-1H-II-FM-7-15-41

Figure 7-13. Cruise chart, clean configuration, +15°C, 16000 feet, (Sheet 41 of 61)



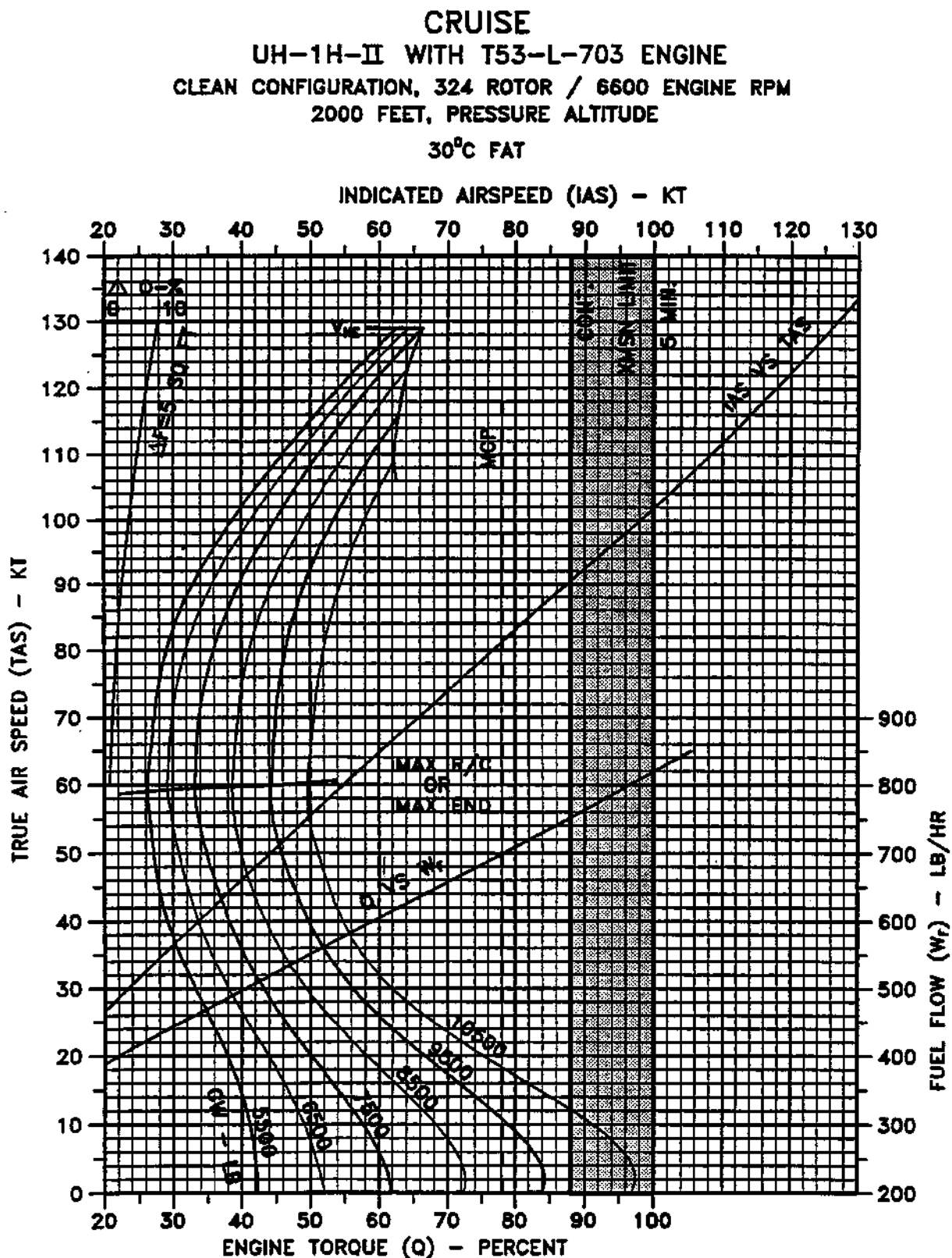
UH-1H-II-FM-7-15-42

Figure 7-13. Cruise chart, clean configuration +15°C, 18000 feet (Sheet 42 of 61)



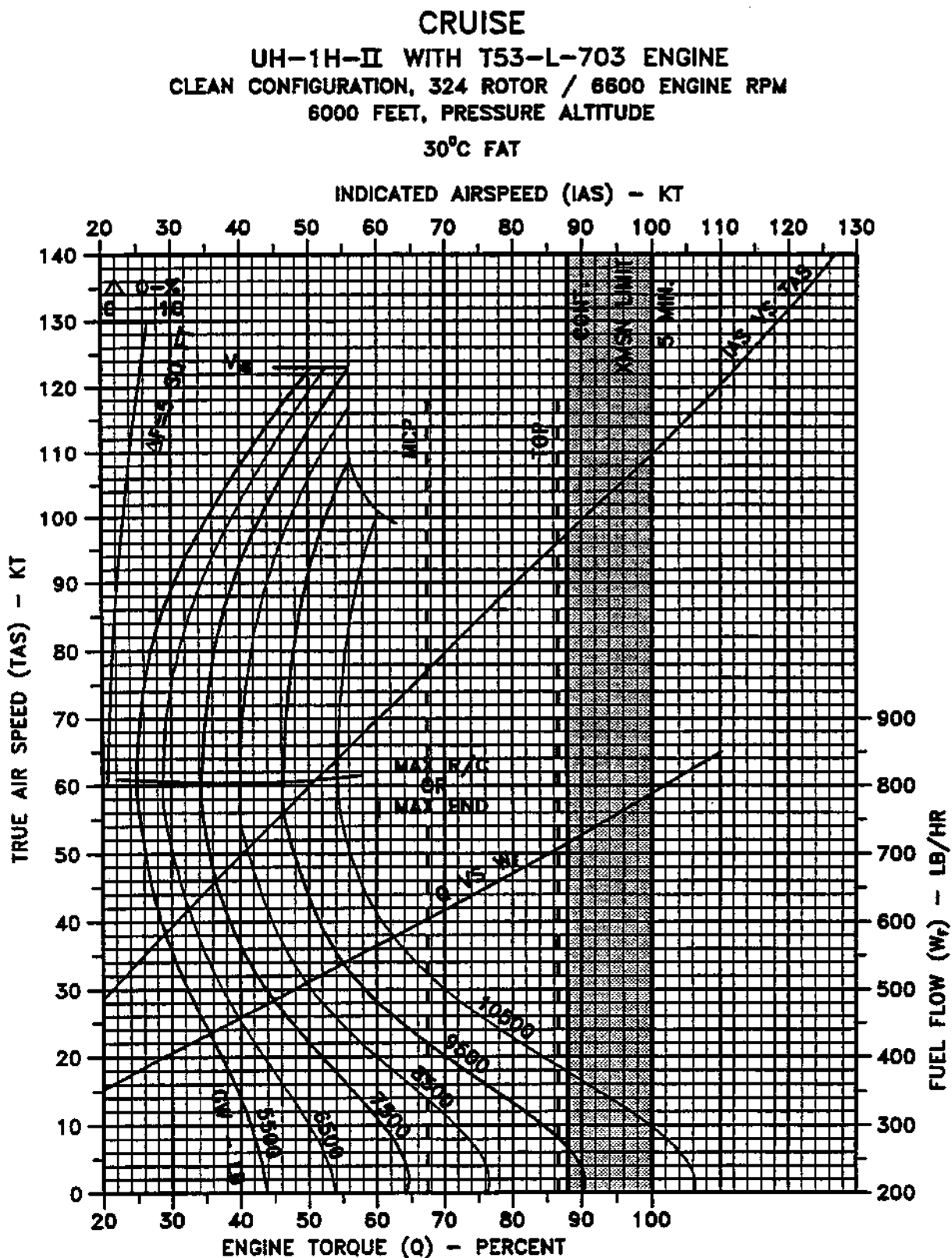
UH-1H-II-FM-7-15-43

Figure 7-13. Cruise chart, clean configuration, 30°C, sea level, (Sheet 43 of 61)



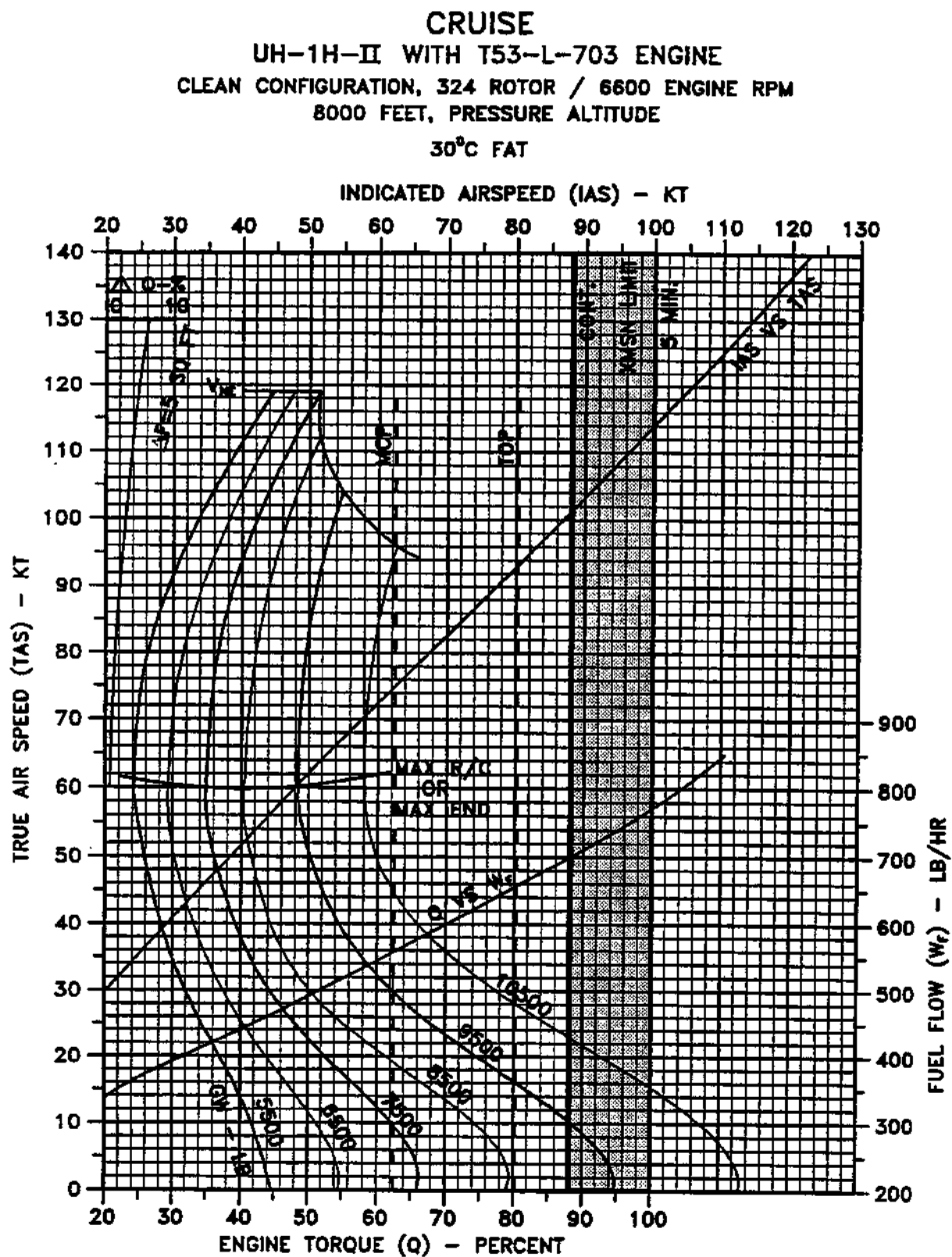
UH-1H-II-FM-7-15-44

Figure 7-13. Cruise chart, clean configuration, 30°C, 2000 feet, (Sheet 44 of 61)



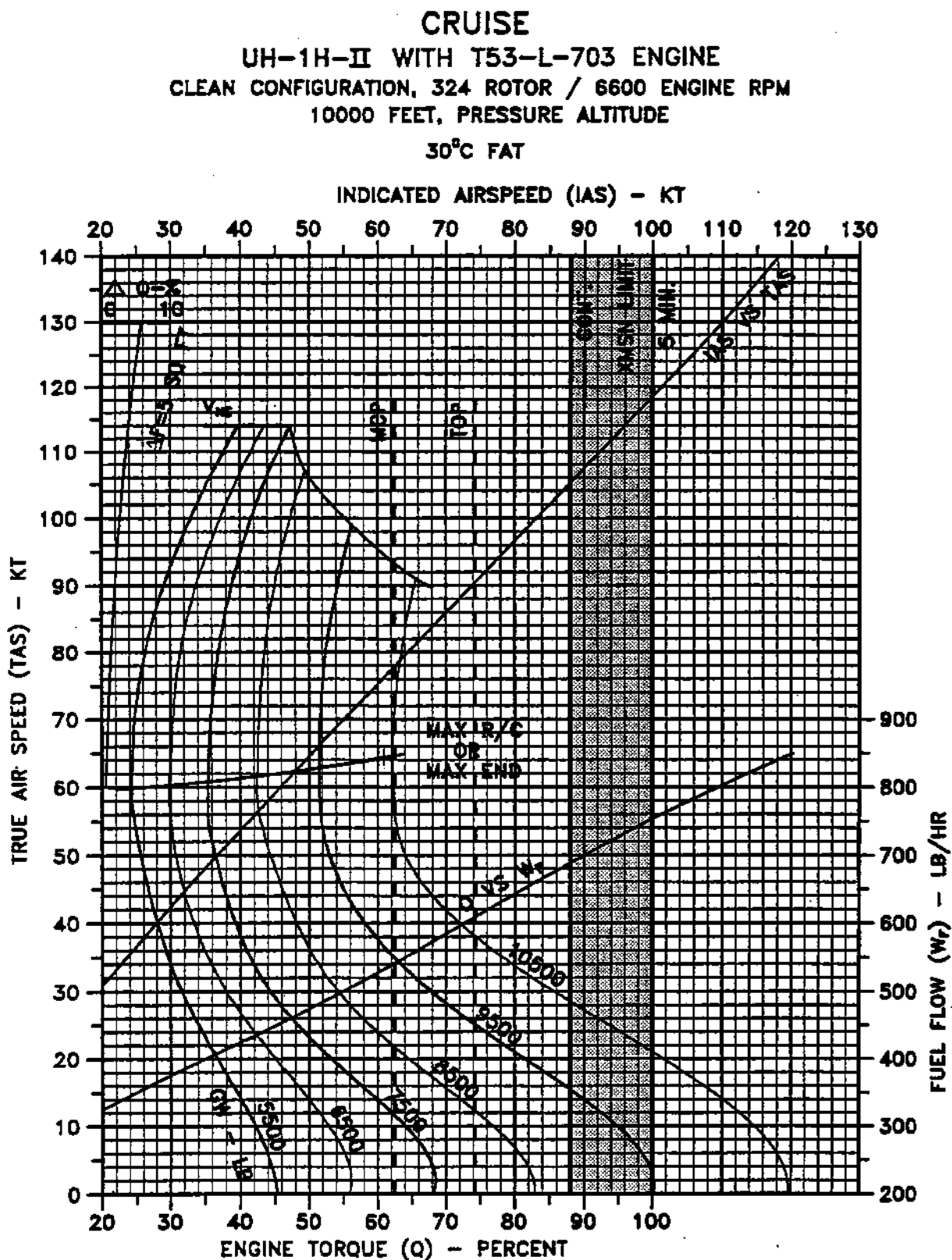
UH-1H-II-FM-7-15-46

Figure 7-13. Cruise chart, clean configuration, 30°C, 6000 feet, (Sheet 46 of 61)



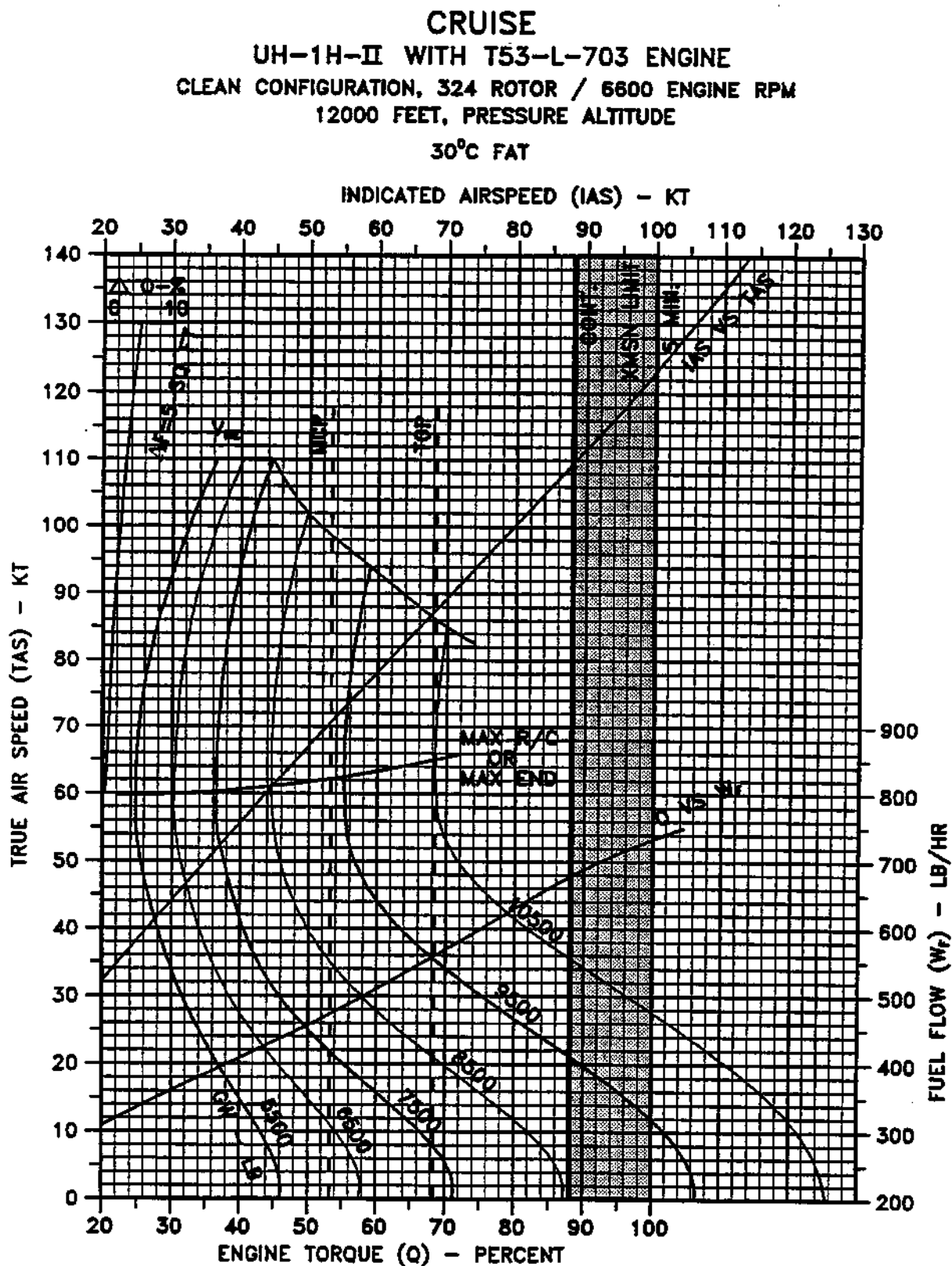
UH-1H-II-FM-7-15-47

Figure 7-13. Cruise chart, clean configuration, 30°C, 8000 feet, (Sheet 47 of 61)



UH-1H-II-FM-7-15-48

Figure 7-13. Cruise chart, clean configuration, 30°C, 10000 feet, (Sheet 48 of 61)

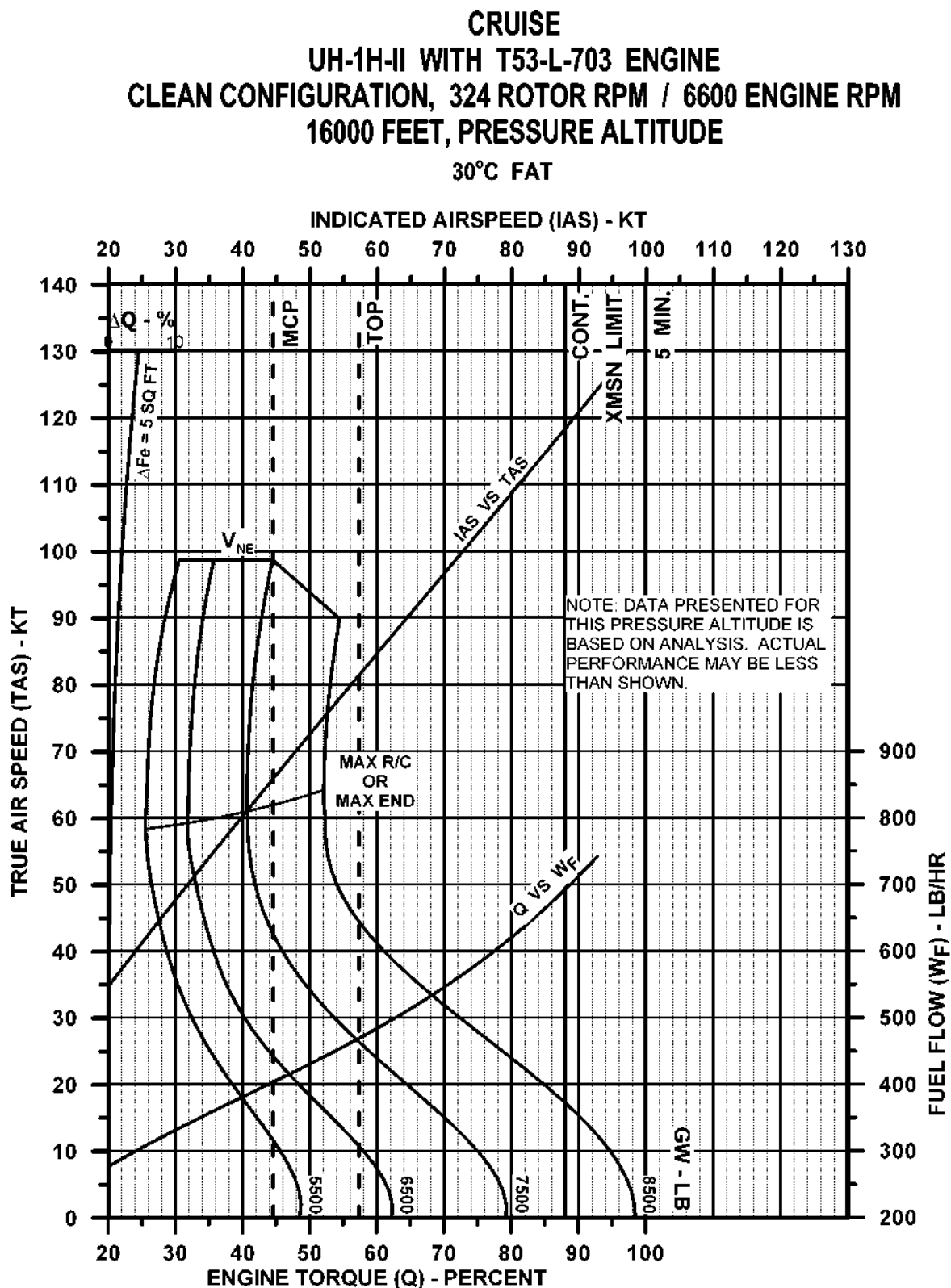


UH-1H-II-FM-7-15-49

Figure 7-13. Cruise chart, clean configuration, 30°C, 12000 feet, (Sheet 49 of 61)



7-00-00
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UH-1H-II-FM-7-15-51

Figure 7-13. Cruise chart, clean configuration, 30°C, 16000 feet, (Sheet 51 of 61)

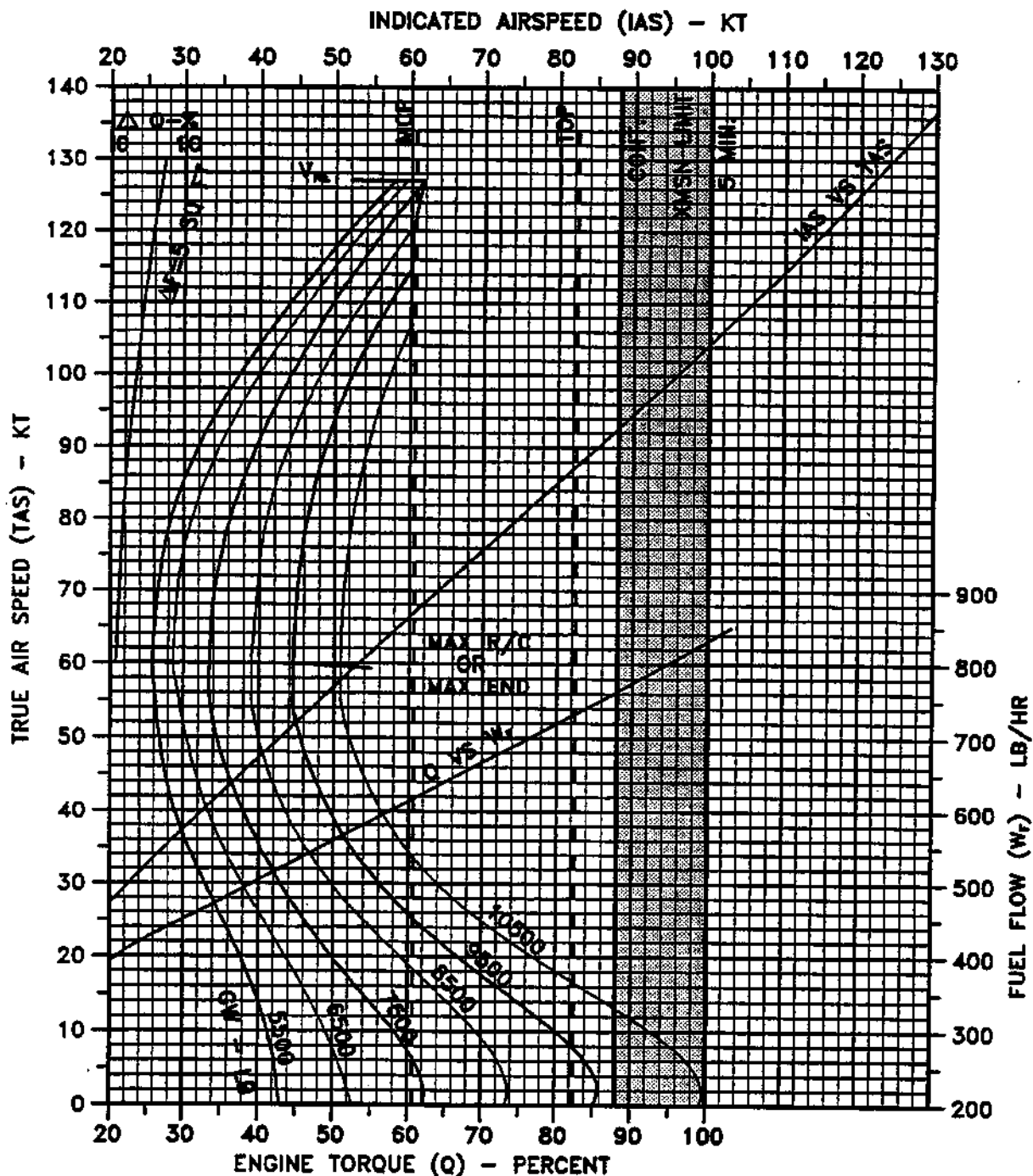
CRUISE

UH-1H-II WITH T53-L-703 ENGINE

CLEAN CONFIGURATION, 324 ROTOR / 6600 ENGINE RPM

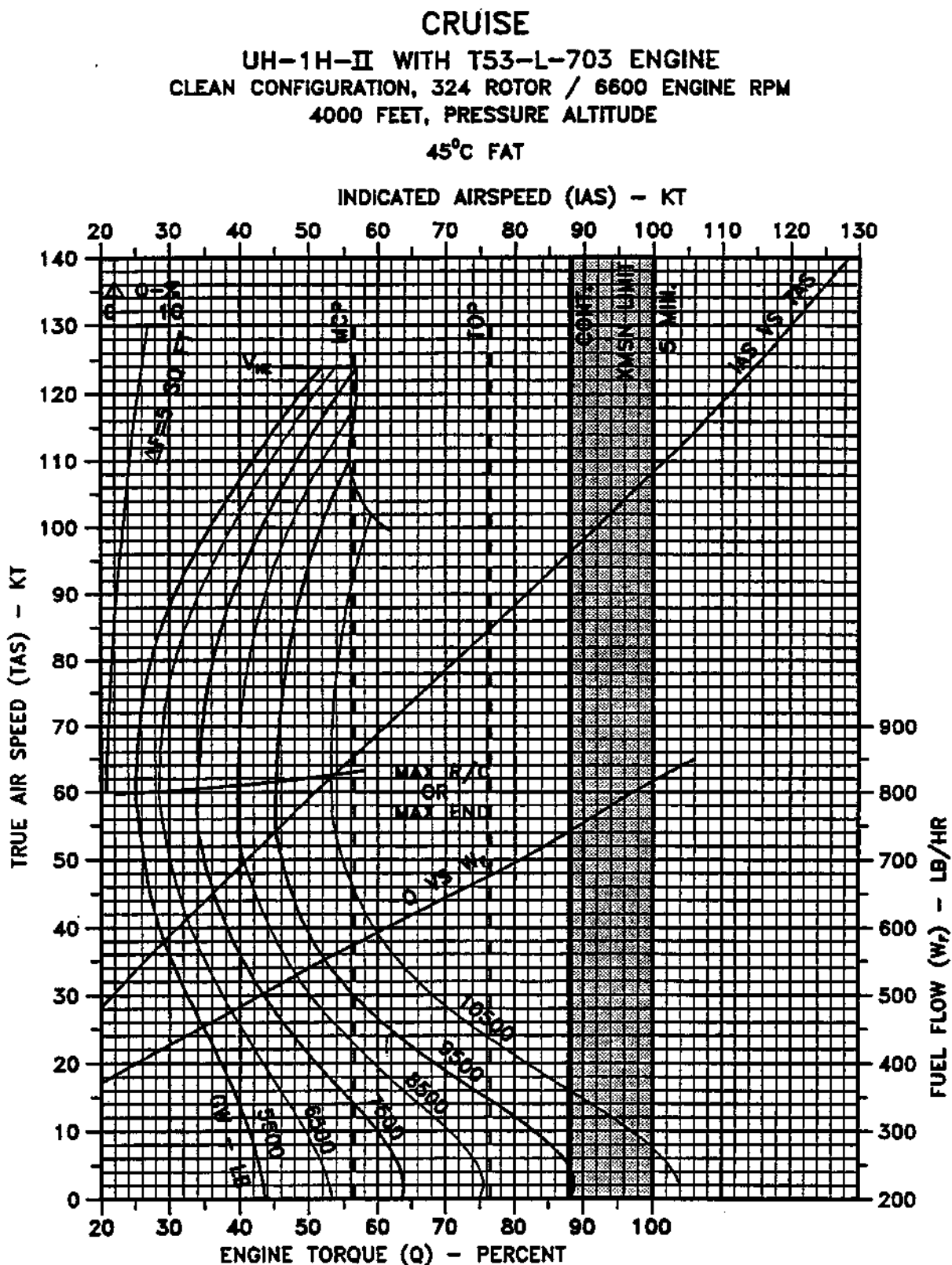
2000 FEET, PRESSURE ALTITUDE

45°C FAT



UH-1H-II-FM-7-15-53

Figure 7-13. Cruise chart, clean configuration, 45°C, 2000 feet, (Sheet 53 of 61)



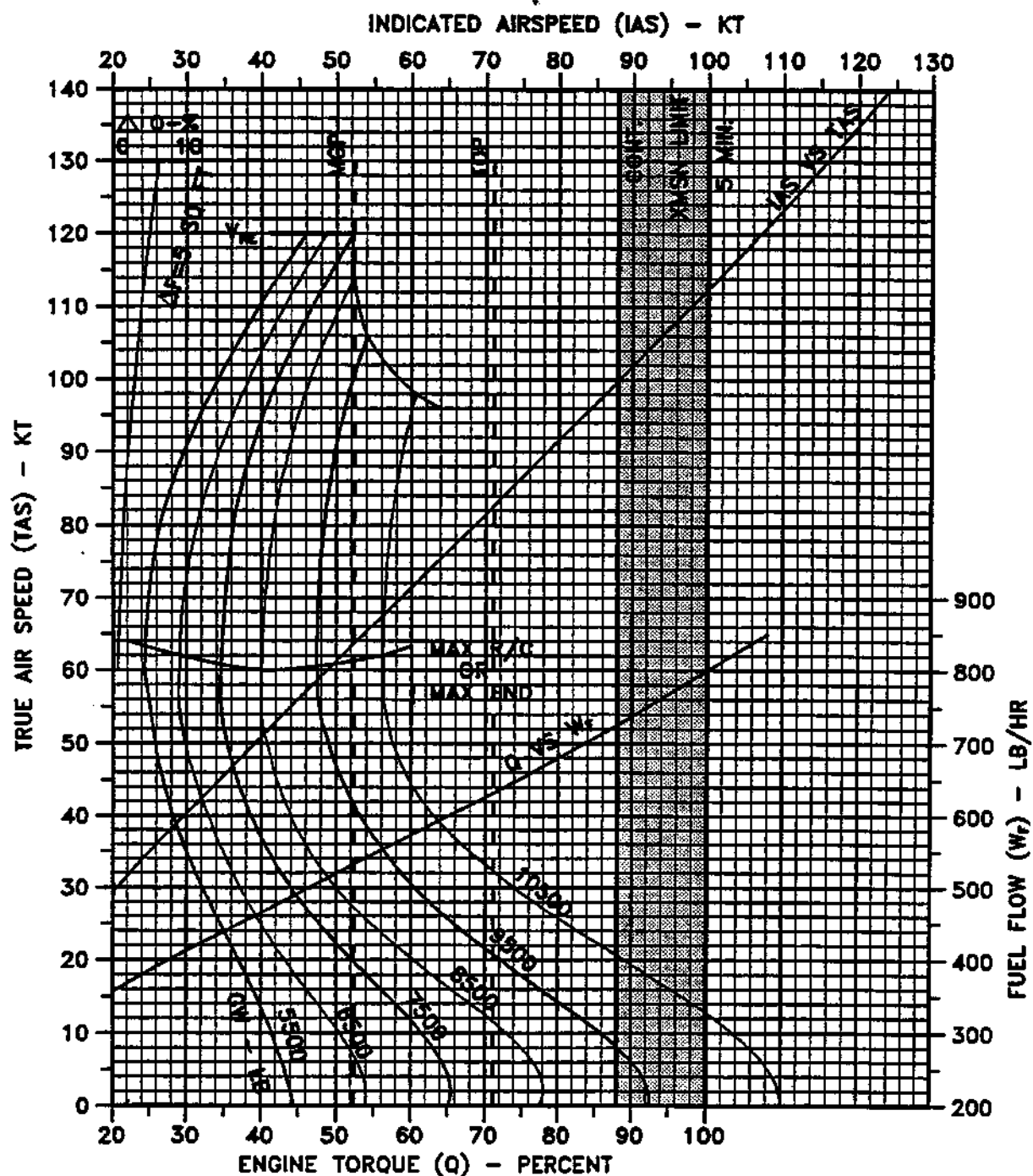
UH-1H-II-FM-7-15-54

Figure 7-13. Cruise chart, clean configuration, 45°C, 4000 feet, (Sheet 54 of 61)

CRUISE

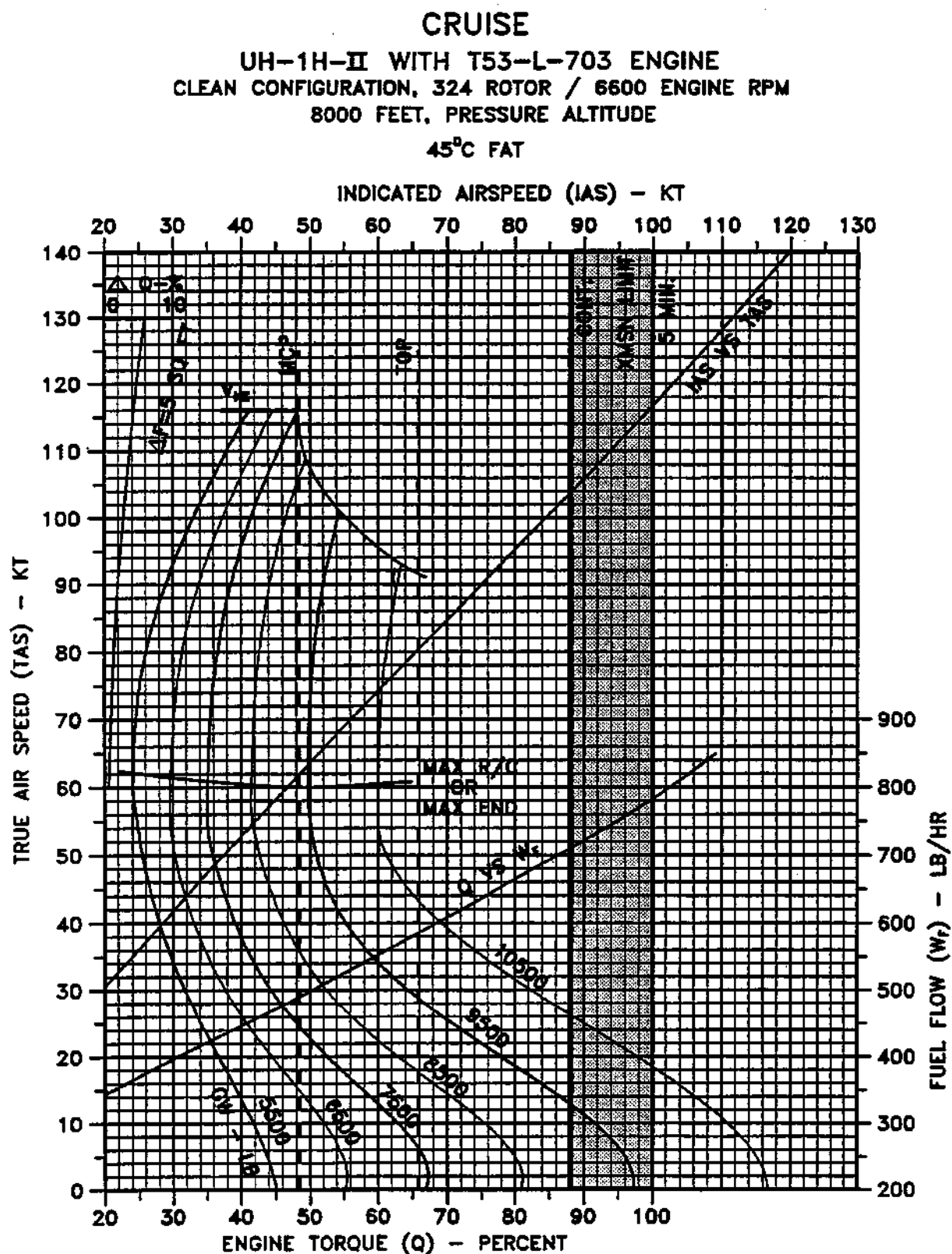
UH-1H-II WITH T53-L-703 ENGINE
 CLEAN CONFIGURATION, 324 ROTOR / 6600 ENGINE RPM
 6000 FEET, PRESSURE ALTITUDE

45°C FAT



UH-1H-II-FM-7-15-55

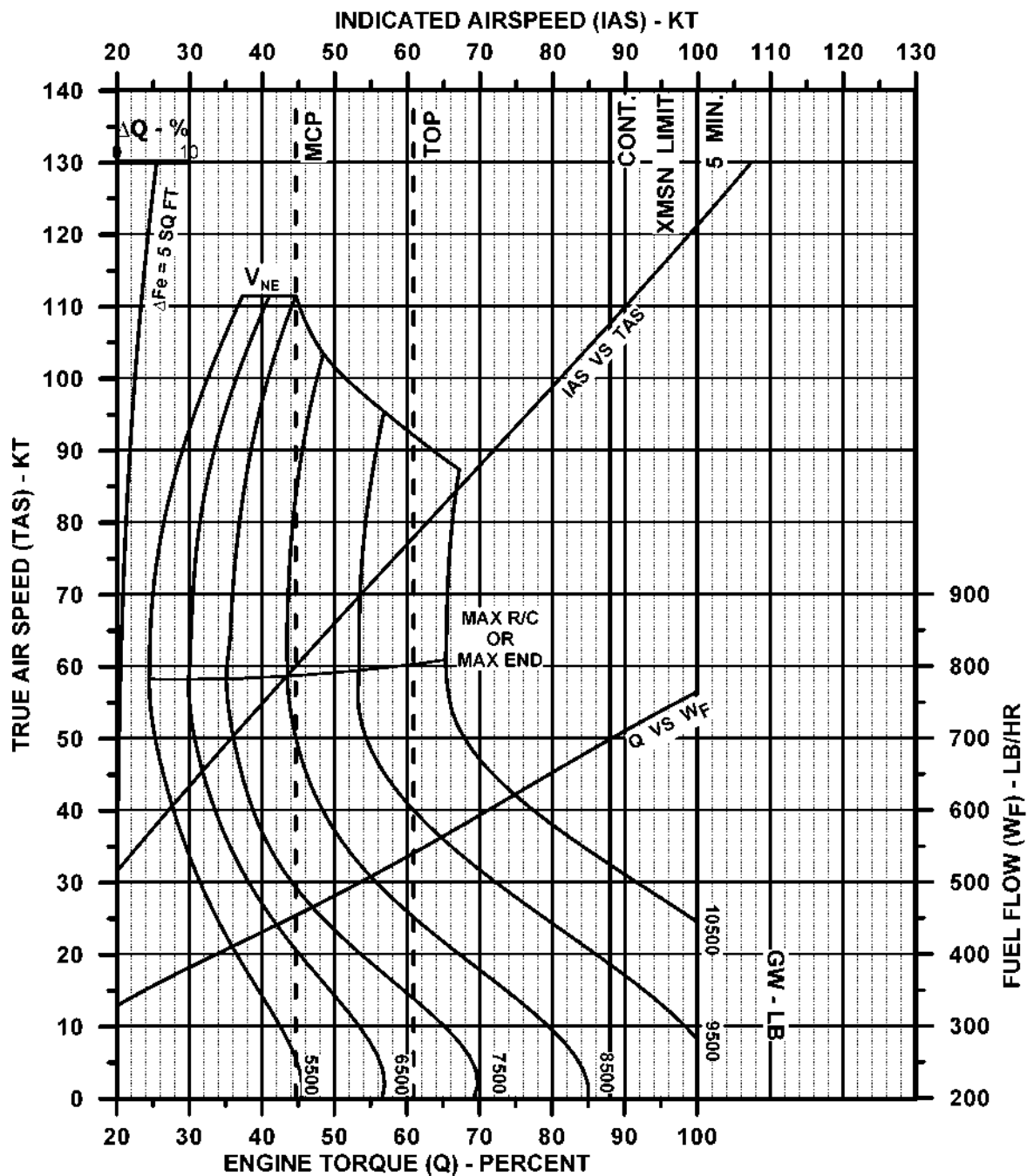
Figure 7-13. Cruise chart, clean configuration, 45°C, 6000 feet, (Sheet 55 of 61)



UH-1H-II-FM-7-15-57

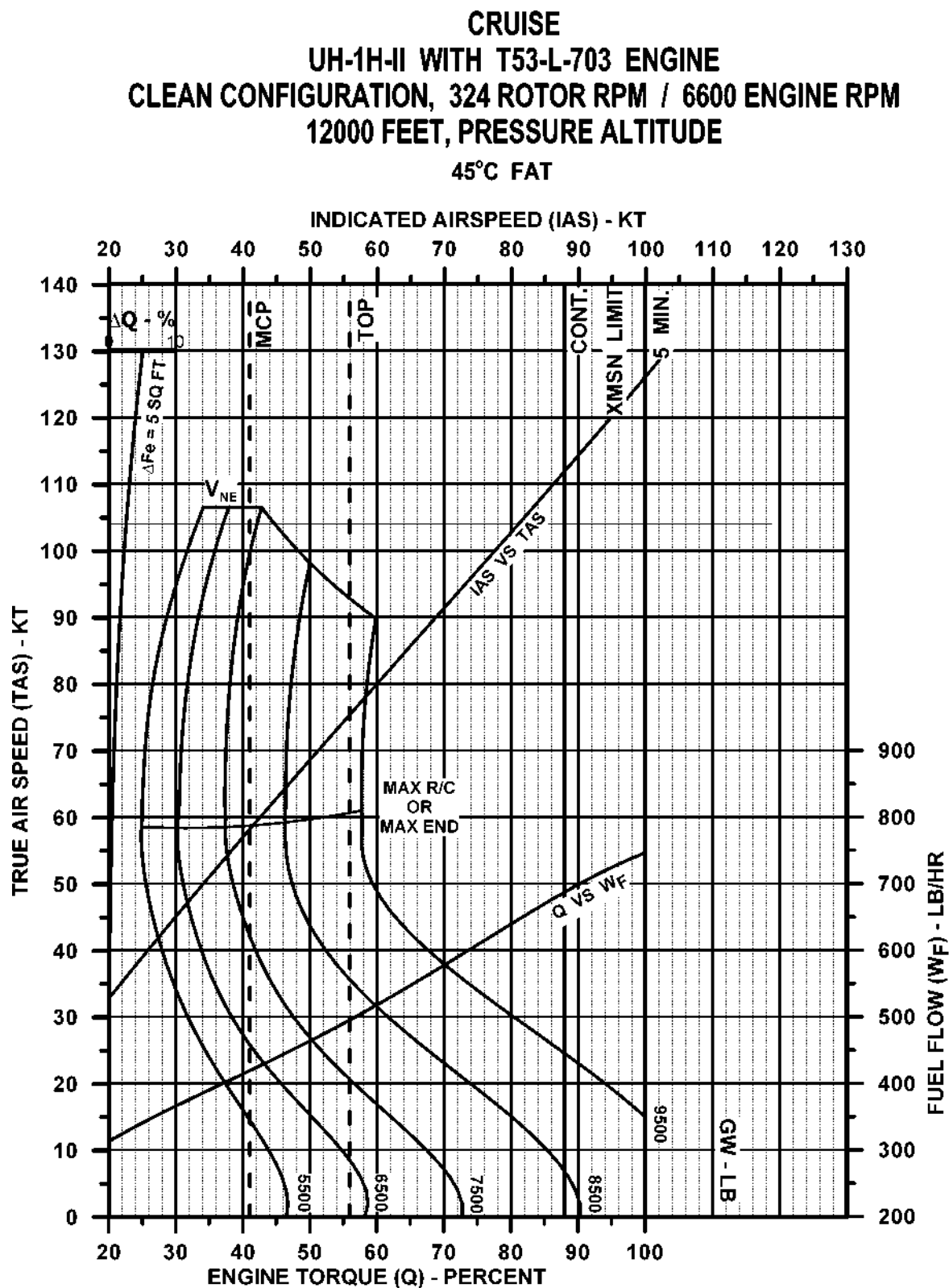
Figure 7-13. Cruise chart, clean configuration, 45°C, 8000 feet, (Sheet 56 of 61)

CRUISE
UH-1H-II WITH T53-L-703 ENGINE
CLEAN CONFIGURATION, 324 ROTOR RPM / 6600 ENGINE RPM
10000 FEET, PRESSURE ALTITUDE
45°C FAT



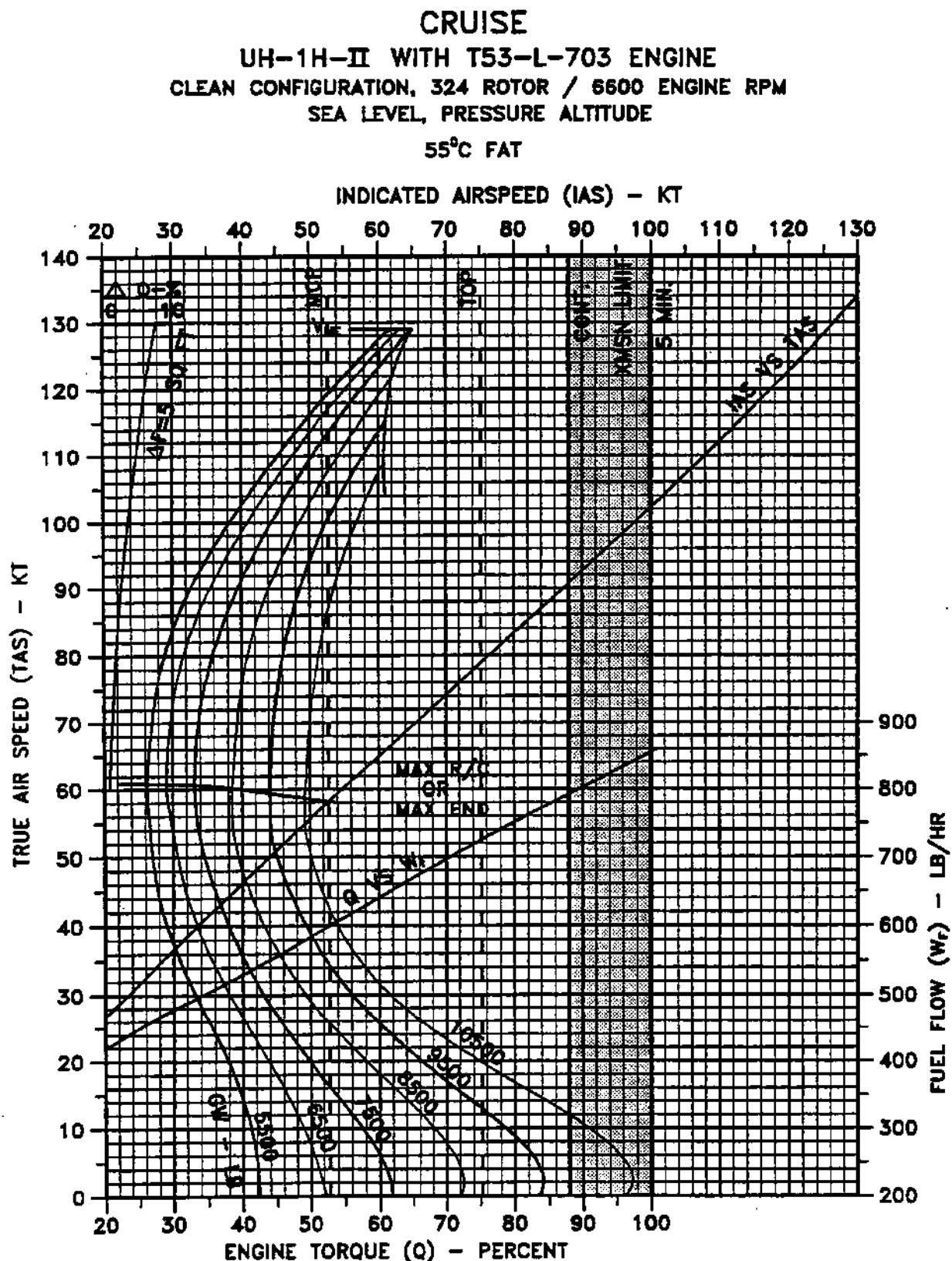
UH-1H-II-FM-7-15-57

Figure 7-13. Cruise chart, clean configuration, 45°C, 10000 feet, (Sheet 57 of 61)



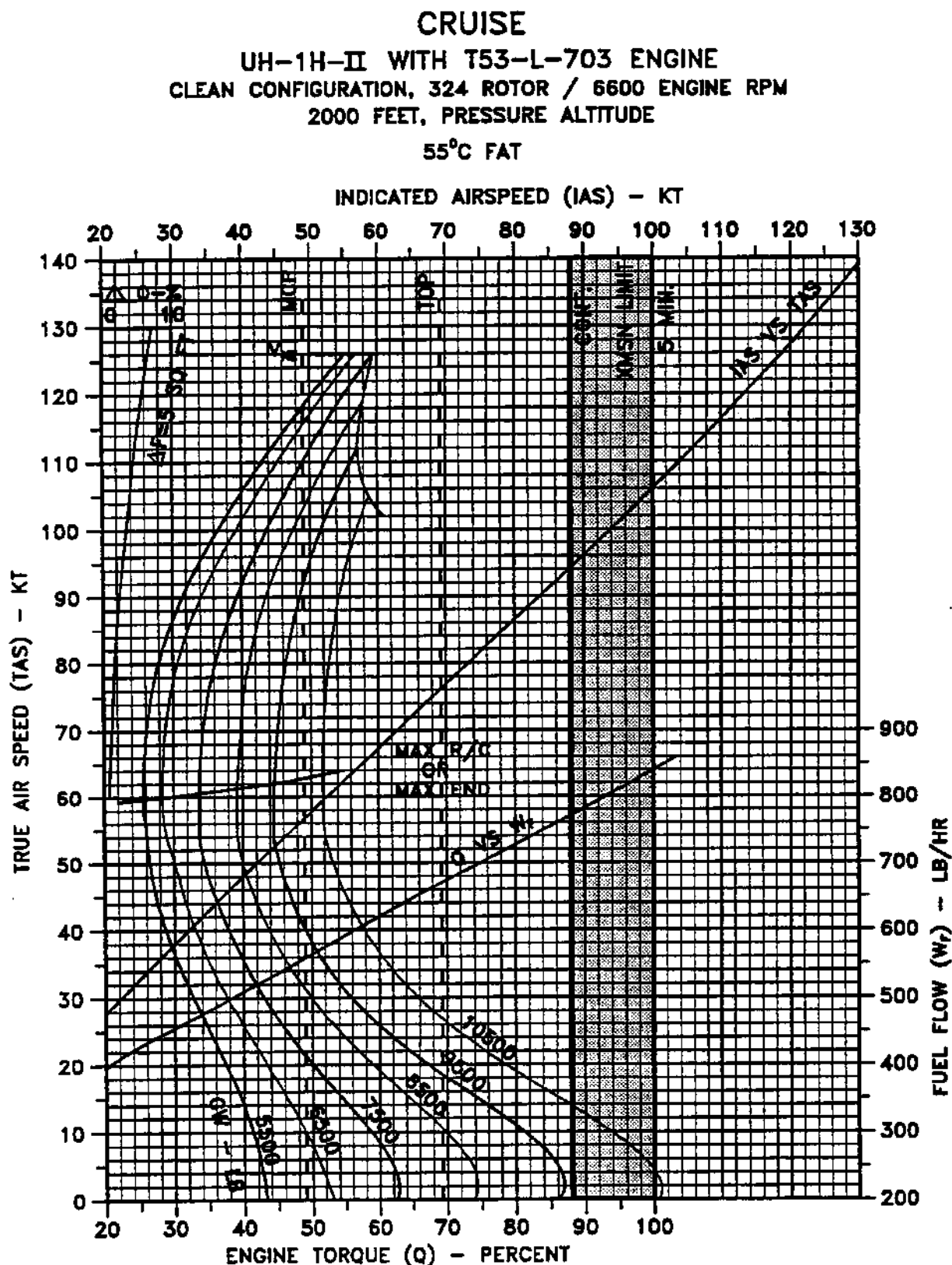
UH-1H-II-FM-7-15-58

Figure 7-13. Cruise chart, clean configuration, 45°C, 12000 feet, (Sheet 58 of 61)



UH-1H-II-FM-7-15-59

Figure 7-13. Cruise chart, clean configuration, 55°C, sea level, (Sheet 59 of 61)

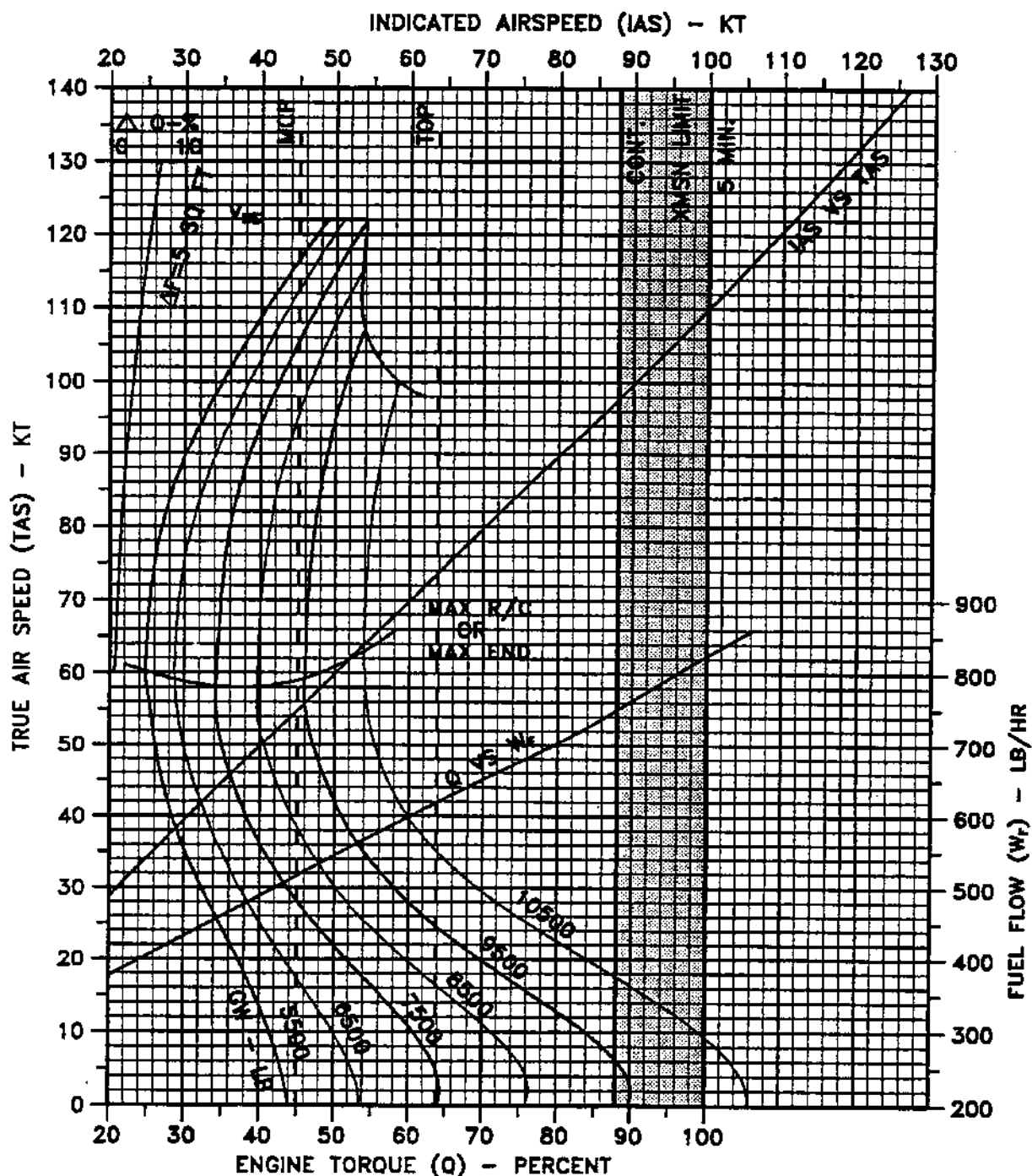


UH-1H-II-FM-7-15-60

Figure 7-13. Cruise chart, clean configuration, 55°C, 2000 feet, (Sheet 60 of 61)

CRUISE

UH-1H-II WITH T53-L-703 ENGINE
 CLEAN CONFIGURATION, 324 ROTOR / 6600 ENGINE RPM
 4000 FEET, PRESSURE ALTITUDE
 55°C FAT



UH-1H-II-FM-7-15-61

Figure 7-13. Cruise chart, clean configuration, 55°C, 4000 feet, (Sheet 61 of 61)